

- [MIMETYPE Function \(Fulltext\)](#)
- [PLAINTEXT Function \(Fulltext\)](#)
- [SCORE Function \(Fulltext\)](#)

Hierarchy Functions

Hierarchy functions allow you to work with hierarchical data such as tables with rows arranged in a tree or directed graph.

SAP HANA supports three types of hierarchy functions: **generator**, **navigation**, and **scalar**.

Following is a list of the supported hierarchy functions, which you can find documentation for in the *SAP HANA Hierarchy Developer Guide* along with complete documentation for the SAP HANA hierarchy feature.

Generator functions

- HIERARCHY Generator Function
- HIERARCHY_LEVELLED Generator Function
- HIERARCHY_SPANTREE Generator Function
- HIERARCHY_TEMPORAL Generator Function

Navigation functions

- HIERARCHY_ANCESTORS Navigation Function
- HIERARCHY_ANCESTORS_AGGREGATE Navigation Function
- HIERARCHY_DESCENDANTS Navigation Function
- HIERARCHY_DESCENDANTS_AGGREGATE Navigation Function
- HIERARCHY_SIBLINGS Navigation Function

Scalar functions

- HIERARCHY_COMPOSITE_ID Scalar Function

Related Information

[SAP HANA Hierarchy Developer Guide](#)

[HIERARCHY Generator Function](#)

[HIERARCHY_ANCESTORS Navigation Function](#)

[HIERARCHY_ANCESTORS_AGGREGATE Navigation Function](#)

[HIERARCHY_COMPOSITE_ID Scalar Function](#)

[HIERARCHY_DESCENDANTS Navigation Function](#)

[HIERARCHY_DESCENDANTS_AGGREGATE Navigation Function](#)

[HIERARCHY_LEVELLED Generator Function](#)

[HIERARCHY_SIBLINGS Navigation Function](#)

[HIERARCHY_SPANTREE Generator Function](#)

[HIERARCHY_TEMPORAL Generator Function](#)

JSON Functions

JSON functions are functions that return or operate on JSON data.

Supported functions:

- [JSON_QUERY Function \(JSON\)](#)
- [JSON_TABLE Function \(JSON\)](#)
- [JSON_VALUE Function \(JSON\)](#)

Miscellaneous Functions

SAP HANA supports many functions that return system values and perform various operations on values, expressions, and return values of other functions.

Supported functions:

- [ALLOW_PRECISION_LOSS Function \(Miscellaneous\)](#)
- [COALESCE Function \(Miscellaneous\)](#)
- [CONVERT_CURRENCY Function \(Miscellaneous\)](#)
- [CONVERT_UNIT Function \(Miscellaneous\)](#)
- [CURRENT_CONNECTION Function \(Miscellaneous\)](#)
- [CURRENT_IDENTITY_VALUE Function \(Miscellaneous\)](#)
- [CURRENT_OBJECT_SCHEMA Function \(Miscellaneous\)](#)
- [CURRENT_SCHEMA Function \(Miscellaneous\)](#)
- [CURRENT_SITE_ID Function \(Miscellaneous\)](#)
- [CURRENT_TRANSACTION_ISOLATION_LEVEL Function \(Miscellaneous\)](#)
- [CURRENT_UPDATE_STATEMENT_SEQUENCE Function \(Miscellaneous\)](#)
- [CURRENT_UPDATE_TRANSACTION Function \(Miscellaneous\)](#)
- [CURRENT_USER Function \(Miscellaneous\)](#)
- [CURRENT_USER_ID Function \(Miscellaneous\)](#)
- [EXPRESSION_MACRO Function \(Miscellaneous\)](#)
- [GREATEST Function \(Miscellaneous\)](#)
- [GROUP_SCORE Function \(Miscellaneous\)](#)
- [GROUPING Function \(Miscellaneous\)](#)
- [GROUPING_ID Function \(Miscellaneous\)](#)
- [HASH_MD5 Function \(Miscellaneous\)](#)
- [HASH_SHA256 Function \(Miscellaneous\)](#)

- [IFNULL Function \(Miscellaneous\)](#)
- [LEAST Function \(Miscellaneous\)](#)
- [MAP Function \(Miscellaneous\)](#)
- [NEWUID Function \(Miscellaneous\)](#)
- [NULLIF Function \(Miscellaneous\)](#)
- [RECORD_COMMIT_TIMESTAMP Function \(Miscellaneous\)](#)
- [RECORD_ID Function \(Miscellaneous\)](#)
- [RESULT_CACHE_ID Function \(Miscellaneous\)](#)
- [RESULT_CACHE_REFRESH_TIME Function \(Miscellaneous\)](#)
- [SESSION_CONTEXT Function \(Miscellaneous\)](#)
- [SESSION_USER Function \(Miscellaneous\)](#)
- [SYSUUID Function \(Miscellaneous\)](#)
- [WIDTH_BUCKET Function \(Miscellaneous\)](#)
- [XMLEXTRACT Function \(Miscellaneous\)](#)
- [XMLEXTRACTVALUE Function \(Miscellaneous\)](#)

Numeric Functions

Numeric functions perform mathematical operations on numerical data types or return numeric information.

Number functions take numeric values, or strings with numeric characters, as inputs and return numeric values. When strings with numeric characters are provided as inputs, implicit conversion from a string to a number is performed automatically before results are computed.

Supported functions:

- [ABS Function \(Numeric\)](#)
- [ACOS Function \(Numeric\)](#)
- [ASIN Function \(Numeric\)](#)
- [ATAN Function \(Numeric\)](#)
- [ATAN2 Function \(Numeric\)](#)
- [BITAND Function \(Numeric\)](#)
- [BITCOUNT Function \(Numeric\)](#)
- [BITNOT Function \(Numeric\)](#)
- [BITOR Function \(Numeric\)](#)
- [BITSET Function \(Numeric\)](#)
- [BITUNSET Function \(Numeric\)](#)

- [BITXOR Function \(Numeric\)](#)
- [CEIL Function \(Numeric\)](#)
- [COS Function \(Numeric\)](#)
- [COSH Function \(Numeric\)](#)
- [COT Function \(Numeric\)](#)
- [EXP Function \(Numeric\)](#)
- [FLOOR Function \(Numeric\)](#)
- [LN Function \(Numeric\)](#)
- [LOG Function \(Numeric\)](#)
- [MOD Function \(Numeric\)](#)
- [NDIV0 Function \(Numeric\)](#)
- [POWER Function \(Numeric\)](#)
- [RAND Function \(Numeric\)](#)
- [RAND_SECURE Function \(Numeric\)](#)
- [ROUND Function \(Numeric\)](#)
- [SIGN Function \(Numeric\)](#)
- [SIN Function \(Numeric\)](#)
- [SINH Function \(Numeric\)](#)
- [SQRT Function \(Numeric\)](#)
- [TAN Function \(Numeric\)](#)
- [TANH Function \(Numeric\)](#)
- [UMINUS Function \(Numeric\)](#)

Security Functions

Security functions provide special functionality for security purposes.

Supported functions:

- [ENCRYPTION_ROOT_KEYS_EXTRACT_KEYS Function \(Security\)](#)
- [ENCRYPTION_ROOT_KEYS_HAS_BACKUP_PASSWORD Function \(Security\)](#)
- [ESCAPE_DOUBLE_QUOTES Function \(Security\)](#)
- [ESCAPE_SINGLE_QUOTES Function \(Security\)](#)
- [GENERATE_PASSWORD Function \(Security\)](#)
- [IS_SQL_INJECTION_SAFE Function \(Security\)](#)

- [VALIDATE_USERGROUP_CONNECT_RESTRICTION_DETAILS Function \(Security\)](#).
- [VALIDATE_USERGROUP_CONNECT_RESTRICTION Function \(Security\)](#).

Series Data Functions

Series data functions provide special functionality for series data and series tables.

Supported functions:

- [SERIES_DISAGGREGATE Function \(Series Data\)](#).
- [SERIES_ELEMENT_TO_PERIOD Function \(Series Data\)](#).
- [SERIES_GENERATE Function \(Series Data\)](#).
- [SERIES_PERIOD_TO_ELEMENT Function \(Series Data\)](#).
- [SERIES_ROUND Function \(Series Data\)](#).

String Functions

String functions perform extraction and manipulation on strings, or return information about strings.

Supported functions:

- [ABAP_ALPHANUM Function \(String\)](#).
- [ABAP_DF16RAW_TO_SMALLDECIMAL Function \(String\)](#).
- [ABAP_DF34RAW_TO_DECIMAL Function \(String\)](#).
- [ABAP_LOWER Function \(String\)](#).
- [ABAP_NUMC Function \(String\)](#).
- [ABAP_UPPER Function \(String\)](#).
- [ASCII Function \(String\)](#).
- [BINTOHEX Function \(String\)](#).
- [BINTONHEX Function \(String\)](#).
- [BINTOSTR Function \(String\)](#).
- [CHAR Function \(String\)](#).
- [CONCAT Function \(String\)](#).
- [CONCAT_NAZ Function \(String\)](#).
- [HAMMING_DISTANCE Function \(String\)](#).
- [HEXTOBIN Function \(String\)](#).
- [HEXTONUM Function \(String\)](#).

- [INITCAP Function \(String\)](#).
- [LCASE Function \(String\)](#).
- [LEFT Function \(String\)](#).
- [LENGTH Function \(String\)](#).
- [LOCATE Function \(String\)](#).
- [LOCATE REGEXPR Function \(String\)](#).
- [LOWER Function \(String\)](#).
- [LPAD Function \(String\)](#).
- [LTRIM Function \(String\)](#).
- [NCHAR Function \(String\)](#).
- [NORMALIZE Function \(String\)](#).
- [NUMTOHEX Function \(String\)](#).
- [OCCURRENCES REGEXPR Function \(String\)](#).
- [REPLACE Function \(String\)](#).
- [REPLACE REGEXPR Function \(String\)](#).
- [RIGHT Function \(String\)](#).
- [RPAD Function \(String\)](#).
- [RTRIM Function \(String\)](#).
- [SOUNDEX Function \(String\)](#).
- [STRTOBIN Function \(String\)](#).
- [SUBSTR AFTER Function \(String\)](#).
- [SUBSTR BEFORE Function \(String\)](#).
- [SUBSTRING Function \(String\)](#).
- [SUBSTRING REGEXPR Function \(String\)](#).
- [TRIM Function \(String\)](#).
- [UCASE Function \(String\)](#).
- [UNICODE Function \(String\)](#).
- [UPPER Function \(String\)](#).
- [XMLTABLE Function \(String\)](#).

Window Functions and the Window Specification

Window functions allow you to perform analytic operations over a set of input rows.

Syntax

This is custom documentation. For more information, please visit [SAP Help Portal](#).

<function_name> **<window_specification>**

<window_specification> ::= OVER ({ [**<window_partition_by_clause>**] [**<window_order_by_clause>**] |

Syntax Elements

<function_name>

Specifies the window function to operate over the data. See the individual function topics for more information on their syntax.

```
<function_name> ::=
BINNING
| CUBIC_SPLINE_APPROX
| CUME_DIST
| DENSE_RANK
| LAG
| LEAD
| LINEAR_APPROX
| NTILE
| PERCENT_RANK
| PERCENTILE_CONT
| PERCENTILE_DISC
| RANDOM_PARTITION
| RANK
| ROW_NUMBER
| SERIES_FILTER
| WEIGHTED_AVG
| <window_aggregate_functions>
| <spatial_functions>
```

For **<window_aggregate_functions>**, see [Window Aggregate Functions](#).

For **<spatial_functions>**, see the *SAP HANA Spatial Reference* guide.

<window_partition_by_clause>

<window_partition_by_clause> ::= PARTITION BY **<expression>** [{ , **<expression>** } ...]

Creates one or more window partitions using the supplied expressions.

<window_order_by_clause>

Defines the sort order of the window.

<window_order_by_clause> ::= ORDER BY **<window_order_by_expression>** [{, **<window_order_by_expression>**

<window_order_by_expression> ::=
<column> [**<collate_clause>**] [ASC | DESC] [NULLS { FIRST | LAST }]

<collate_clause> ::= COLLATE **<collation_name>**

<collate_clause>

Specifies the collation to use. **<collate_clause>** can only be used on columns defined as NVARCHAR or VARCHAR.

<collation_name> is one of the supported collation names listed in the COLLATIONS system view.

[ASC | DESC]

ASC sorts records in ascending order and DESC sorts records in descending order. The default is ASC.

[NULLS FIRST | NULLS LAST]

Specifies the ordering of NULL values.

<position>

<position> uses the entries in the select list to define the ordering required. For example:

```
SELECT col1, col2 FROM t ORDER BY 2;
```

ORDER BY 2 indicates that ordering should be undertaken using the second expression in the select list, which in this case is col2.

<window_frame_clause>

Defines the rows (bounds) of a window partition by specifying the number of rows preceding or following with respect to the current row. If you specify a <window_frame_clause>, you must also specify a <window_order_by_clause>.

```
<window_frame_clause> ::= <window_frame_unit> <window_frame_extent>
```

```
<window_frame_extent> ::= { <window_frame_start> | <window_frame_between> }
```

```
<window_frame_unit> ::= ROWS
```

```
<window_frame_start> ::= { UNBOUNDED PRECEDING | <window_frame_preceding> | CURRENT ROW }
```

```
<window_frame_preceding> ::= <unsigned_integer> PRECEDING
```

```
<window_frame_between> ::= BETWEEN <lower_window_frame_bound> AND <upper_window_frame_bound>
```

```
<lower_window_frame_bound> ::= <window_frame_bound>
```

```
<upper_window_frame_bound> ::= <window_frame_bound>
```

```
<window_frame_bound> := { <window_frame_start> | UNBOUNDED FOLLOWING | <window_frame_following> }
```

```
<window_frame_following> ::= <unsigned_integer> FOLLOWING
```

<window_frame_between> specifies the lower (starting) bound and the upper (ending) bound of the window frame. The upper bound cannot be smaller than the lower bound.

UNBOUND PRECEDING specifies that the window starts at the first row of the partition. It can only be specified as <window_frame_start>. CURRENT ROW specifies that the window starts or ends at the current row when used with ROWS.

PRECEDING indicates that the number of rows or values to precede the current row. <unsigned_integer> should be 0 or a positive integer literal.

UNBOUNDED FOLLOWING specifies that the window ends at the last row of the partition. UNBOUNDED FOLLOWING can only be specified as a window frame end.

FOLLOWING indicates the number of rows or values to follow the current row. When specified as the window frame start, the window frame end should be <unsigned_integer> FOLLOWING. The <unsigned_integer> should be 0 or a positive integer literal.

Description

The window functions allow you to divide the result sets of a query, or a logical partition of a query, into groups of rows called window partitions.

A window partition is specified by one or more expressions in the OVER clause.

Window functions not in the following list must have an ORDER BY clause in the OVER clause:

- AVG
- CORR
- CORR_SPEARMAN
- COUNT
- MAX
- MEDIAN

- MIN
- PERCENTILE_CONT
- PERCENTILE_DISC
- ROW_NUMBER
- STDDEV
- SUM
- VAR

Result sets are first partitioned as specified by PARTITION BY clause, and then sorted by ORDER BY clause specification within the window partition.

Finally window functions are applied to each row within window partition boundaries.

The ORDER BY clause in the OVER clause is only used to evaluate window function so that the order of resulting rows is non-deterministic if not specified by ORDER BY for SELECT.

The default window frame of the window function depends on whether or not a window ORDER BY clause is specified. If a window ORDER BY clause is specified, the default window frame becomes 'between UNBOUNDED PRECEDING and CURRENT ROW'. That is to say that the window function computes on rows preceding or peer with current row. As a result, the function returns cumulative values.

If a window ORDER BY clause is not specified, the default window frame becomes 'between UNBOUNDED PRECEDING and UNBOUNDED FOLLOWING'. That is to say that the window function computes on the whole window partition. As a result, the function returns the same value within a window partition regardless of current row.

Examples

The examples use the example table T defined below.

```
CREATE TABLE myTable (Class CHAR(10), Value INT, Offset INT);
INSERT INTO myTable VALUES('A', 1, 1);
INSERT INTO myTable VALUES('A', 3, 3);
INSERT INTO myTable VALUES('A', 5, NULL);
INSERT INTO myTable VALUES('A', 5, 2);
INSERT INTO myTable VALUES('A', 10, 0);
INSERT INTO myTable VALUES('B', 1, 3);
INSERT INTO myTable VALUES('B', 1, 1);
INSERT INTO myTable VALUES('B', 7, 1);

SELECT Class, VAL,
       ROW_NUMBER() OVER (PARTITION BY Class ORDER BY Value) AS ROW_NUM,
       RANK() OVER (PARTITION BY Class ORDER BY Value) AS RANK,
       DENSE_RANK() OVER (PARTITION BY Class ORDER BY Value) AS DENSE_RANK
FROM myTable;
```

class	val	row_num	rank	dense_rank
A	1	1	1	1
A	3	2	2	2
A	5	3	3	3

A	5	4	3	3
A	10	5	5	4
B	1	1	1	1
B	1	2	1	1
B	7	3	3	2

Related Information

[Window Aggregate Functions](#)

[Spatial Functions](#)

[BINNING Function \(Window\)](#)

[CUBIC_SPLINE_APPROX Function \(Window\)](#)

[COLLATIONS System View](#)

[CUME_DIST Function \(Window\)](#)

[DENSE_RANK Function \(Window\)](#)

[LAG Function \(Window\)](#)

[LEAD Function \(Window\)](#)

[LINEAR_APPROX Function \(Window\)](#)

[NTILE Function \(Window\)](#)

[PERCENT_RANK Function \(Window\)](#)

[PERCENTILE_CONT Function \(Window\)](#)

[PERCENTILE_DISC Function \(Window\)](#)

[RANDOM_PARTITION Function \(Window\)](#)

[RANK Function \(Window\)](#)

[ROW_NUMBER Function \(Window\)](#)

[SERIES_FILTER Function \(Window\)](#)

[WEIGHTED_AVG Function \(Window\)](#)

[SAP HANA Spatial Reference](#)

Window Aggregate Functions

Some aggregate functions can be used as window functions over a window specification.

Syntax

```
<aggregate_function_or_expression_as_window_function> ::=
  <aggregate_function_name> ( <arguments> ) <window_specification>
```

```
<aggregate_function_name> ::=
  AVG
  | CORR
  | CORR_SPEARMAN
  | COUNT
  | FIRST_VALUE
  | LAST_VALUE
  | MAX
  | MEDIAN
  | MIN
  | NTH_VALUE
  | STDDEV
```

Syntax Elements

<aggregate_function_name>

Specifies the aggregate function or aggregate expression to use as a window function.

<arguments>

Specifies the input to the function. Refer to the individual function topic for the definition of *<arguments>*.

<window_specification>

Specifies the window definition. See [Window Functions and the Window Specification](#)

Description

Window aggregate functions are aggregate functions that can be used as window functions.

Examples

```
CREATE ROW TABLE T (class CHAR(10), val INT, offset INT);
INSERT INTO T VALUES('A', 1, 1);
INSERT INTO T VALUES('A', 3, 3);
INSERT INTO T VALUES('A', 5, null);
INSERT INTO T VALUES('A', 5, 2);
INSERT INTO T VALUES('A', 10, 0);
INSERT INTO T VALUES('B', 1, 3);
INSERT INTO T VALUES('B', 1, 1);
INSERT INTO T VALUES('B', 7, 1);

SELECT class,
       val,
       offset,
       COUNT(*) OVER (PARTITION BY class) AS c1,
       COUNT(offset) OVER (PARTITION BY class) AS c2,
       COUNT(*) OVER (PARTITION BY class ORDER BY val) AS c3,
       COUNT(offset) OVER (PARTITION BY class ORDER BY val) AS c4,
       MAX(val) OVER (PARTITION BY class) AS m1,
       MAX(val) OVER (PARTITION BY class ORDER BY val) AS m2
FROM T;
```

CLASS	VAL	OFFSET	C1	C2	C3	C4	M1	M2
A	1	1	5	4	1	1	10	1
A	3	3	5	4	2	2	10	3
A	5	null	5	4	4	3	10	5
A	5	2	5	4	4	3	10	5
A	10	0	5	4	5	4	10	10
B	1	3	3	3	2	2	7	1
B	1	1	3	3	2	2	7	1
B	7	1	3	3	3	3	7	7

Related Information

[Window Functions and the Window Specification](#)

[AVG Function \(Aggregate\)](#)

[CORR Function \(Aggregate\)](#)

[CORR_SPEARMAN Function \(Aggregate\)](#)

[COUNT Function \(Aggregate\)](#)

[FIRST_VALUE Function \(Aggregate\)](#)

[LAST_VALUE Function \(Aggregate\)](#)

[MAX Function \(Aggregate\)](#)

[MEDIAN Function \(Aggregate\)](#)

[MIN Function \(Aggregate\)](#)

[NTH_VALUE Function \(Aggregate\)](#)

[STDDEV Function \(Aggregate\)](#)

[SUM Function \(Aggregate\)](#)

[VAR Function \(Aggregate\)](#)

SQL Statements

SAP HANA supports many SQL statements to allow you to perform such tasks as create database objects, administer your system, and manipulate data.

[Alphabetical List of Statements](#)

[Access Control Statements](#)

Access control statements enable database administrators to create, alter, and drop access to the SAP HANA database.

[Backup and Recovery Statements](#)

These statements allow users to perform backup and recovery operations.

[Data Definition Statements](#)

Data definition statements define structures in the SAP HANA database.

[Data Import Export Statements](#)

The following statements import and export data to and from the SAP HANA database.

[Data Manipulation Statements](#)

The following statements enable you to manipulate data within the SAP HANA database.

[Client-side Encryption Statements](#)

Client-side encryption statements enable database administrators to administer client-side encrypted data in the SAP HANA database.

[JSON Document Store Statements](#)

There are several JSON Document Store-specific variants of common DDL and DML SQL statements.

[Procedural Statements](#)

Procedural statements manage system and user-defined procedures for the SAP HANA database.

[Session Management Statements](#)

The following SQL statements manage database sessions.

[System Management Statements](#)

The following statements enable you to manage system configuration settings in your SAP HANA database.

[Tenant Database Management Statements](#)

The following SQL statements manage SAP HANA tenant databases.

[Transaction Management Statements](#)

The following SQL statements manage transactions in the SAP HANA database.

The following SQL statements manage workload classes and mappings.

Access Control Statements

Access control statements enable database administrators to create, alter, and drop access to the SAP HANA database.

Supported Statements

- [ALTER AUDIT POLICY Statement \(Access Control\)](#)
- [ALTER CREDENTIAL Statement \(Access Control\)](#)
- [ALTER JWT PROVIDER Statement \(Access Control\)](#)
- [ALTER LDAP PROVIDER Statement \(Access Control\)](#)
- [ALTER REMOTE SOURCE Statement \(Access Control\)](#)
- [ALTER ROLE Statement \(Access Control\)](#)
- [ALTER SAML PROVIDER Statement \(Access Control\)](#)
- [ALTER STRUCTURED PRIVILEGE Statement \(Access Control\)](#)
- [ALTER USER Statement \(Access Control\)](#)
- [ALTER USERGROUP Statement \(Access Control\)](#)
- [CREATE AUDIT POLICY Statement \(Access Control\)](#)
- [CREATE CREDENTIAL Statement \(Access Control\)](#)
- [CREATE JWT PROVIDER Statement \(Access Control\)](#)
- [CREATE LDAP PROVIDER Statement \(Access Control\)](#)
- [CREATE REMOTE SOURCE Statement \(Access Control\)](#)
- [CREATE ROLE Statement \(Access Control\)](#)
- [CREATE SAML PROVIDER Statement \(Access Control\)](#)
- [CREATE STRUCTURED PRIVILEGE Statement \(Access Control\)](#)
- [CREATE USER Statement \(Access Control\)](#)
- [CREATE USERGROUP Statement \(Access Control\)](#)
- [DROP AUDIT POLICY Statement \(Access Control\)](#)
- [DROP CREDENTIAL Statement \(Access Control\)](#)
- [DROP JWT PROVIDER Statement \(Access Control\)](#)
- [DROP LDAP PROVIDER Statement \(Access Control\)](#)
- [DROP REMOTE SOURCE Statement \(Access Control\)](#)
- [DROP ROLE Statement \(Access Control\)](#)
- [DROP SAML PROVIDER Statement \(Access Control\)](#)

- [DROP STRUCTURED PRIVILEGE Statement \(Access Control\)](#)
- [DROP USER Statement \(Access Control\)](#)
- [DROP USERGROUP Statement \(Access Control\)](#)
- [GRANT Statement \(Access Control\)](#)
- [REVOKE Statement \(Access Control\)](#)
- [VALIDATE LDAP PROVIDER Statement \(Access Control\)](#)
- [VALIDATE USER Statement \(Access Control\)](#)

Backup and Recovery Statements

These statements allow users to perform backup and recovery operations.

Supported Statements

- [BACKUP CANCEL Statement \(Backup and Recovery\)](#)
- [BACKUP CATALOG DELETE Statement \(Backup and Recovery\)](#)
- [BACKUP CATALOG RETAINED Statement \(Backup and Recovery\)](#)
- [BACKUP CHECK Statement \(Backup and Recovery\)](#)
- [BACKUP CHECK ACCESS Statement \(Backup and Recovery\)](#)
- [BACKUP DATA Statement \(Backup and Recovery\)](#)
- [BACKUP DATA CLOSE SNAPSHOT Statement \(Backup and Recovery\)](#)
- [BACKUP DATA CREATE SNAPSHOT Statement \(Backup and Recovery\)](#)
- [BACKUP DATA DROP SNAPSHOT Statement \(Backup and Recovery\)](#)
- [BACKUP ENCRYPTION ROOT KEYS Statement \(Backup and Recovery\)](#)
- [BACKUP LIST DATA Statement \(Backup and Recovery\)](#)
- [RECOVER DATA Statement \(Backup and Recovery\)](#)
- [RECOVER DATABASE Statement \(Backup and Recovery\)](#)
- [RECOVER ENCRYPTION ROOT KEYS Statement \(Backup and Recovery\)](#)

Data Definition Statements

Data definition statements define structures in the SAP HANA database.

Supported Statements

- [ALTER FULLTEXT INDEX Statement \(Data Definition\)](#)
- [ALTER INDEX Statement \(Data Definition\)](#)

- [ALTER SEQUENCE Statement \(Data Definition\)](#)
- [ALTER STATISTICS Statement \(Data Definition\)](#)
- [ALTER SCHEDULER JOB Statement \(Data Definition\)](#)
- [ALTER TABLE Statement \(Data Definition\)](#)
- [ALTER VIEW Statement \(Data Definition\)](#)
- [ALTER VIRTUAL TABLE Statement \(Data Definition\)](#)
- [ANNOTATE Statement \(Data Definition\)](#)
- [COMMENT ON Statement \(Data Definition\)](#)
- [CREATE FULLTEXT INDEX Statement \(Data Definition\)](#)
- [CREATE GRAPH WORKSPACE Statement \(Data Definition\)](#)
- [CREATE INDEX Statement \(Data Definition\)](#)
- [CREATE FUZZY SEARCH INDEX Statement \(Data Definition\)](#)
- [CREATE PROJECTION VIEW Statement \(Data Definition\)](#)
- [CREATE SCHEDULER JOB Statement \(Data Definition\)](#)
- [CREATE SCHEMA Statement \(Data Definition\)](#)
- [CREATE SCHEMA SYNONYM Statement \(Data Definition\)](#)
- [CREATE SEQUENCE Statement \(Data Definition\)](#)
- [CREATE STATISTICS Statement \(Data Definition\)](#)
- [CREATE SYNONYM Statement \(Data Definition\)](#)
- [CREATE TABLE Statement \(Data Definition\)](#)
- [CREATE TRIGGER Statement \(Data Definition\)](#)
- [CREATE VIEW Statement \(Data Definition\)](#)
- [CREATE VIRTUAL TABLE Statement \(Data Definition\)](#)
- [DROP FULLTEXT INDEX Statement \(Data Definition\)](#)
- [DROP GRAPH WORKSPACE Statement \(Data Definition\)](#)
- [DROP INDEX Statement \(Data Definition\)](#)
- [DROP SCHEDULER JOB Statement \(Data Definition\)](#)
- [DROP SCHEMA Statement \(Data Definition\)](#)
- [DROP SCHEMA SYNONYM Statement \(Data Definition\)](#)
- [DROP SEQUENCE Statement \(Data Definition\)](#)
- [DROP STATISTICS Statement \(Data Definition\)](#)
- [DROP SYNONYM Statement \(Data Definition\)](#)
- [DROP TABLE Statement \(Data Definition\)](#)
- [DROP TRIGGER Statement \(Data Definition\)](#)

- [DROP VIEW Statement \(Data Definition\)](#)
- [REFRESH STATISTICS Statement \(Data Definition\)](#)
- [REFRESH VIEW Statement \(Data Definition\)](#)
- [RENAME COLUMN Statement \(Data Definition\)](#)
- [RENAME INDEX Statement \(Data Definition\)](#)
- [RENAME SCHEMA Statement \(Data Definition\)](#)
- [RENAME TABLE Statement \(Data Definition\)](#)

Data Import Export Statements

The following statements import and export data to and from the SAP HANA database.

Supported Statements

- [EXPORT INTO Statement \(Data Import Export\)](#)
- [EXPORT Statement \(Data Import Export\)](#)
- [IMPORT FROM Statement \(Data Import Export\)](#)
- [IMPORT SCAN Statement \(Data Import Export\)](#)
- [IMPORT Statement \(Data Import Export\)](#)

Data Manipulation Statements

The following statements enable you to manipulate data within the SAP HANA database.

Supported Statements

- [DELETE Statement \(Data Manipulation\)](#)
- [EXPLAIN PLAN Statement \(Data Manipulation\)](#)
- [INSERT Statement \(Data Manipulation\)](#)
- [LOAD Statement \(Data Manipulation\)](#)
- [MERGE DELTA Statement \(Data Manipulation\)](#)
- [MERGE INTO Statement \(Data Manipulation\)](#)
- [REPLACE Statement \(Data Manipulation\)](#)
- [SELECT Statement \(Data Manipulation\)](#)
- [TRUNCATE TABLE Statement \(Data Manipulation\)](#)
- [UNLOAD Statement \(Data Manipulation\)](#)
- [UPDATE Statement \(Data Manipulation\)](#)

- [UPSERT Statement \(Data Manipulation\)](#)

Client-side Encryption Statements

Client-side encryption statements enable database administrators to administer client-side encrypted data in the SAP HANA database.

Supported Statements

- [ALTER CLIENTSIDE ENCRYPTION COLUMN KEY Statement \(Encryption\)](#)
- [ALTER CLIENTSIDE ENCRYPTION KEYPAIR Statement \(Encryption\)](#)
- [CREATE CLIENTSIDE ENCRYPTION COLUMN KEY Statement \(Encryption\)](#)
- [CREATE CLIENTSIDE ENCRYPTION KEYPAIR Statement \(Encryption\)](#)
- [DROP CLIENTSIDE ENCRYPTION COLUMN KEY Statement \(Encryption\)](#)
- [DROP CLIENTSIDE ENCRYPTION KEYPAIR Statement \(Encryption\)](#)

JSON Document Store Statements

There are several JSON Document Store-specific variants of common DDL and DML SQL statements.

Supported Statements

To see the list of statements and how to use them to create collection tables and manage JSON data, see the *SAP HANA JSON Document Store Guide*.

Related Information

[SAP HANA JSON Document Store Guide](#)

[M_TABLES System View](#)

[TABLES System View](#)

Procedural Statements

Procedural statements manage system and user-defined procedures for the SAP HANA database.

Supported Statements

- [ALTER FUNCTION Statement \(Procedural\)](#)
- [ALTER PROCEDURE Statement \(Procedural\)](#)
- [CALL Statement \(Procedural\)](#)
- [CREATE FUNCTION Statement \(Procedural\)](#)
- [CREATE PROCEDURE Statement \(Procedural\)](#)

- [CREATE TYPE Statement \(Procedural\)](#)
- [CREATE VIRTUAL FUNCTION Statement \(Procedural\)](#)
- [CREATE VIRTUAL PROCEDURE Statement \(Procedural\)](#)
- [DO BEGIN...END Statement \(Procedural\)](#)
- [DROP FUNCTION Statement \(Procedural\)](#)
- [DROP PROCEDURE Statement \(Procedural\)](#)
- [DROP TYPE Statement \(Procedural\)](#)

Session Management Statements

The following SQL statements manage database sessions.

Supported Statements

- [CONNECT Statement \(Session Management\)](#)
- [SET \[SESSION\] Statement \(Session Management\)](#)
- [SET HISTORY SESSION Statement \(Session Management\)](#)
- [SET SCHEMA Statement \(Session Management\)](#)
- [UNSET \[SESSION\] Statement \(Session Management\)](#)

Related Information

[Session Variables](#)

System Management Statements

The following statements enable you to manage system configuration settings in your SAP HANA database.

Supported Statements

- [ALTER PSE Statement \(System Management\)](#)
- [ALTER SYSTEM {ADD | ALTER | REMOVE} STATEMENT HINT Statement \(System Management\)](#)
- [ALTER SYSTEM {ADD | ACTIVATE | UPDATE | DROP} KEY MANAGEMENT CONFIGURATION Statement \(System Management\)](#)
- [ALTER SYSTEM {ADD | REMOVE} ABSTRACT SQL PLAN FILTER \(System Management\)](#)
- [ALTER SYSTEM ALTER CONFIGURATION Statement \(System Management\)](#)
- [ALTER SYSTEM ALTER DATAVOLUME ADD PARTITION Statement \(System Management\)](#)
- [ALTER SYSTEM ALTER DATAVOLUME DROP PARTITION Statement \(System Management\)](#)
- [ALTER SYSTEM ALTER SESSION SET Statement \(System Management\)](#)

- [ALTER SYSTEM ALTER SESSION UNSET Statement \(System Management\)](#)
- [ALTER SYSTEM ALTER TABLE PLACEMENT Statement \(System Management\)](#)
- [ALTER SYSTEM APPLICATION ENCRYPTION Statement \(System Management\)](#)
- [ALTER SYSTEM BACKUP ENCRYPTION Statement \(System Management\)](#)
- [ALTER SYSTEM CANCEL \[WORK IN\] SESSION Statement \(System Management\)](#)
- [ALTER SYSTEM CLEAR AUDIT LOG Statement \(System Management\)](#)
- [ALTER SYSTEM CLEAR CACHE Statement \(System Management\)](#)
- [ALTER SYSTEM CLEAR COLUMN JOIN DATA STATISTICS \(System Management\)](#)
- [ALTER SYSTEM CLEAR INIFILE CONTENT HISTORY Statement \(System Management\)](#)
- [ALTER SYSTEM CLEAR RESULT CACHE Statement \(System Management\)](#)
- [ALTER SYSTEM CLEAR SQL PLAN CACHE Statement \(System Management\)](#)
- [ALTER SYSTEM CLEAR TIMEZONE CACHE Statement \(System Management\)](#)
- [ALTER SYSTEM CLEAR TRACES Statement \(System Management\)](#)
- [ALTER SYSTEM CLIENTPKI { UPDATE / DROP } \(System Management\)](#)
- [ALTER SYSTEM CREATE RUNTIMEDUMP Statement \(System Management\)](#)
- [ALTER SYSTEM CREATE WAITGRAPH Statement \(System Management\)](#)
- [ALTER SYSTEM DISCONNECT SESSION Statement \(System Management\)](#)
- [ALTER SYSTEM {ENABLE | DISABLE | REMOVE} ABSTRACT SQL PLAN \(System Management\)](#)
- [ALTER SYSTEM {ENABLE | DISABLE} ALL \[ASYNCHRONOUS | SYNCHRONOUS\] TABLE REPLICAS Statement \(System Management\)](#)
- [ALTER SYSTEM {ENABLE | DISABLE} STATEMENT HINT Statement \(System Management\)](#)
- [ALTER SYSTEM {ENABLE | DISABLE} SYSTEM REPLICATION Statement \(System Management\)](#)
- [ALTER SYSTEM ENCRYPTION CONFIGURATION Statement \(System Management\)](#)
- [ALTER SYSTEM LOAD PERFTRACE Statement \(System Management\)](#)
- [ALTER SYSTEM LOG ENCRYPTION Statement \(System Management\)](#)
- [ALTER SYSTEM LOGGING Statement \(System Management\)](#)
- [ALTER SYSTEM MIGRATE ABSTRACT SQL PLAN \(System Management\)](#)
- [ALTER SYSTEM PERSISTENCE ENCRYPTION Statement \(System Management\)](#)
- [ALTER SYSTEM {PIN | UNPIN} SQL PLAN CACHE ENTRY Statement \(System Management\)](#)
- [ALTER SYSTEM RECLAIM DATA SPACE Statement \(System Management\)](#)
- [ALTER SYSTEM RECLAIM DATAVOLUME Statement \(System Management\)](#)
- [ALTER SYSTEM RECLAIM FLEXIBLE TABLES Statement \(System Management\)](#)
- [ALTER SYSTEM RECLAIM LOB SPACE Statement \(System Management\)](#)
- [ALTER SYSTEM RECLAIM LOG Statement \(System Management\)](#)

- [ALTER SYSTEM RECLAIM VERSION SPACE Statement \(System Management\)](#)
- [ALTER SYSTEM RECOMPILE SQL PLAN CACHE ENTRY Statement \(System Management\)](#)
- [ALTER SYSTEM RECONFIGURE SERVICE Statement \(System Management\)](#)
- [ALTER SYSTEM REFRESH RESULT CACHE Statement \(System Management\)](#)
- [ALTER SYSTEM REFRESH RESULT CACHE ENTRY Statement \(System Management\)](#)
- [ALTER SYSTEM {REGISTER | UNREGISTER} SYSTEM REPLICATION SITE Statement \(System Management\)](#)
- [ALTER SYSTEM REMOVE RESULT CACHE ENTRY Statement \(System Management\)](#)
- [ALTER SYSTEM REMOVE SQL PLAN CACHE ENTRY Statement \(System Management\)](#)
- [ALTER SYSTEM REMOVE TRACES Statement \(System Management\)](#)
- [ALTER SYSTEM RESET MONITORING VIEW Statement \(System Management\)](#)
- [ALTER SYSTEM SAVE PERFTRACE Statement \(System Management\)](#)
- [ALTER SYSTEM SAVEPOINT Statement \(System Management\)](#)
- [ALTER SYSTEM SET ENCRYPTION ROOT KEYS BACKUP PASSWORD Statement \(System Management\)](#)
- [ALTER SYSTEM {START | STOP} APPLY ABSTRACT SQL PLAN \(System Management\)](#)
- [ALTER SYSTEM {START | STOP | SAVE | CLEAR} KERNEL PROFILER Statement \(System Management\)](#)
- [ALTER SYSTEM {START | STOP} CAPTURE ABSTRACT SQL PLAN \(System Management\)](#)
- [ALTER SYSTEM {START | STOP | CLEAR} SQLSCRIPT PLAN PROFILER Statement \(System Management\)](#)
- [ALTER SYSTEM {START | STOP} PERFTRACE Statement \(System Management\)](#)
- [ALTER SYSTEM {START | STOP} SQLSCRIPT CODE COVERAGE Statement \(System Management\)](#)
- [ALTER SYSTEM STOP SERVICE Statement \(System Management\)](#)
- [ALTER SYSTEM UPDATE ABSTRACT SQL PLAN Statement \(System Management\)](#)
- [ALTER SYSTEM VALIDATE ENCRYPTION ROOT KEYS BACKUP PASSWORD Statement \(System Management\)](#)
- [CREATE CERTIFICATE Statement \(System Management\)](#)
- [CREATE PSE Statement \(System Management\)](#)
- [CREATE PUBLIC KEY Statement \(System Management\)](#)
- [DROP CERTIFICATE Statement \(System Management\)](#)
- [DROP PSE Statement \(System Management\)](#)
- [DROP PUBLIC KEY Statement \(System Management\)](#)
- [SET PSE Statement \(System Management\)](#)
- [SET SYSTEM LICENSE Statement \(System Management\)](#)
- [UNSET PSE Statement \(System Management\)](#)
- [UNSET SYSTEM LICENSE ALL Statement \(System Management\)](#)

Tenant Database Management Statements

The following SQL statements manage SAP HANA tenant databases.

Supported Statements

- [ALTER DATABASE Statement \(Tenant Database Management\)](#)
- [ALTER SYSTEM START DATABASE Statement \(Tenant Database Management\)](#)
- [ALTER SYSTEM STOP DATABASE Statement \(Tenant Database Management\)](#)
- [CREATE DATABASE Statement \(Tenant Database Management\)](#)
- [DROP DATABASE Statement \(Tenant Database Management\)](#)
- [RENAME DATABASE Statement \(Tenant Database Management\)](#)

Transaction Management Statements

The following SQL statements manage transactions in the SAP HANA database.

Supported Statements

- [COMMIT Statement \(Transaction Management\)](#)
- [LOCK TABLE Statement \(Transaction Management\)](#)
- [RELEASE SAVEPOINT Statement \(Transaction Management\)](#)
- [ROLLBACK Statement \(Transaction Management\)](#)
- [ROLLBACK TO SAVEPOINT Statement \(Transaction Management\)](#)
- [SAVEPOINT Statement \(Transaction Management\)](#)
- [SET TRANSACTION Statement \(Transaction Management\)](#)
- [SET TRANSACTION AUTOCOMMIT DDL Statement \(Transaction Management\)](#)

Workload Management Statements

The following SQL statements manage workload classes and mappings.

Supported Statements

- [ALTER WORKLOAD CLASS Statement \(Workload Management\)](#)
- [ALTER WORKLOAD MAPPING Statement \(Workload Management\)](#)
- [CREATE WORKLOAD CLASS Statement \(Workload Management\)](#)
- [CREATE WORKLOAD MAPPING Statement \(Workload Management\)](#)
- [DROP WORKLOAD CLASS Statement \(Workload Management\)](#)

- [DROP WORKLOAD MAPPING Statement \(Workload Management\)](#)

System Limitations

Limitations to take into consideration when administering an SAP HANA database.

Aside from the table below, most system limits can also be viewed by querying the M_SYSTEM_LIMITS system view (`SELECT * FROM M_SYSTEM_LIMITS ;`). However, your values might differ depending on the hardware and software configuration your system uses.

For details of sizing and limitations regarding HDI containers, refer to the topic SAP HDI Sizing and Limitations in the guide *SAP HANA Deployment Infrastructure (HDI) Reference for SAP HANA Platform* (see Related Information below).

Limitation Area	Limit	M_SYSTEM_LIMITS view name for the limitation
Database size limit	Row Store: 1,945 GB Column Store: Dependent on size of physical memory	MAXIMUM_SIZE_OF_ROW_STORE
Number of locks	Unlimited for record locks, 16,383 for table locks	MAXIMUM_NUMBER_OF_TABLE_LOCKS
Number of sessions	65,536	MAXIMUM_NUMBER_OF_SESSIONS
Schema Limitations		
Number of schemas per SAP HANA instance	Maximum value of BIGINT data type	
Identifier length	127 bytes	MAXIMUM_LENGTH_OF_IDENTIFIER
Length of an alias name	128 characters	MAXIMUM_LENGTH_OF_ALIAS_NAME
Table name length	Same as Identifier length	MAXIMUM_LENGTH_OF_IDENTIFIER
Column name length	Same as Identifier length	MAXIMUM_LENGTH_OF_IDENTIFIER
Length of a string literal	8 MB	MAXIMUM_LENGTH_OF_STRING_LITERAL
Number of hex characters in a binary literal	8,192 Bytes	MAXIMUM_LENGTH_OF_BINARY_LITERAL
Tables and View Limitations		
Number of columns in a table	64,000 This limit can vary based on context,	MAXIMUM_NUMBER_OF_COLUMNS_IN_TABLE

	for example, in the context of virtual tables, SAP HANA may be limited by the capabilities of the remote system and the limit of the other DBMS may apply instead. In cases such as this, the limit that is met first becomes the actual limit.	
Number of columns in a row table	1,000	MAXIMUM_NUMBER_OF_COLUMNS_IN_ROW_TABLE
Number of columns in a view	64,000	MAXIMUM_NUMBER_OF_COLUMNS_IN_VIEW
Number of rows in each table	Limited by storage size RS: 1,945 GB/sizeof(row), CS: 2,100,000,000 * number of partitions	
Length of a row	Limited by RS storage size (1,945 GB per index server)	
Size of a non-partitioned table	Limited by RS storage size (1,945 GB per index server)	
Number of partitions in a CS table	16,000	MAXIMUM_NUMBER_OF_PARTITIONS_IN_CSTABLE
Number of triggers per table per DML statement	1,024	MAXIMUM_NUMBER_OF_TRIGGERS_PER_TABLE_PER_DML
Number of records per (non-partitioned) table	2,100,000,000	
Indexes and Constraints		
Number of indexes for a table	1,023	MAXIMUM_NUMBER_OF_INDEXES_IN_TABLE
Number of primary key columns in each table	16	MAXIMUM_NUMBER_OF_COLUMNS_IN_PRIMARY_KEY
Number of primary key columns in each column store table	1,000	MAXIMUM_NUMBER_OF_COLUMNS_IN_PRIMARY_KEY_IN_COLUMN_TABLE
Number of columns in an index	16	MAXIMUM_NUMBER_OF_COLUMNS_IN_INDEX
Number of columns in a UNIQUE constraint	16	MAXIMUM_NUMBER_OF_COLUMNS_IN_UNIQUE_CONSTRAINT

Size of sum of primary key, index, UNIQUE constraint	16,384 Bytes	MAXIMUM_SIZE_OF_KEY_IN_INDEX
Number of indexes in row store	256,000	
Number of unique tokens in a fulltext index	4.294.967.295	
SQL		
Length of an SQL statement	2,147,483,648 Bytes	MAXIMUM_LENGTH_OF_SQL_STATEMENT
Depth of SQL view nesting	128	MAXIMUM_DEPTH_OF_SQL_VIEW_NESTING
Maximum depth of SQL parse tree This limitation does not show in M_SYSTEM_LIMITS unless a limit is configured to something other than 0 (no limit) using the max_parse_tree_depth parameter in indexerver.ini.	0 0 (unlimited)	MAXIMUM_DEPTH_OF_SQL_PARSE_TREE
Maximum depth of joins in a statement. This limitation does not show in M_SYSTEM_LIMITS unless a limit is configured to something other than 0 (no limit) using the max_join_depth parameter in indexerver.ini.	0	MAXIMUM_DEPTH_OF_JOINS
Number of columns in an ORDER BY	65,535	MAXIMUM_NUMBER_OF_COLUMNS_IN_ORDER_BY
Number of columns in a GROUP BY	65,535	MAXIMUM_NUMBER_OF_COLUMNS_IN_GROUP_BY
Number of elements in IN predicates	65,535	MAXIMUM_NUMBER_OF_COLUMNS_IN_IN_PREDICATE
Number of elements in SELECT clause	65,535	MAXIMUM_NUMBER_OF_OUTPUT_COLUMNS_IN_STATEMENT
Number of tables in a statement. This limitation does not show in M_SYSTEM_LIMITS unless a limit is configured to something other than 0 (no limit) using the max_table_count_in_statement parameter in indexerver.ini.	0	MAXIMUM_NUMBER_OF_TABLES_IN_STATEMENT
LOB Limitations		

Maximum size of an in-memory LOB for a column store table	2 GB	MAXIMUM_SIZE_OF_MEMORY_LOB_IN_COLUMN_STORE
Maximum size of an in-memory LOB for a row store table	2 GB	MAXIMUM_SIZE_OF_MEMORY_LOB_IN_ROW_STORE
Maximum size of a packed LOB	1,013,760 bytes	MAXIMUM_SIZE_OF_PACKED_LOB
Maximum size of a LOB on disk	4,294,967,295 bytes	MAXIMUM_SIZE_OF_DISK_LOB
Procedures		
Size of all stored procedures	1,945 GB	MAXIMUM_SIZE_OF_ALL_STORED_PROCEDURES
Size of a procedure definition	2 GB	MAXIMUM_SIZE_OF_PROCEDURE_DEFINITION

Related Information

[M_SYSTEM_LIMITS System View](#)

[SAP HDI Sizing and Limitations](#)

SQL Error Codes

Each SAP HANA error is identified by a numeric error code and has a short descriptive text.

The M_ERROR_CODES system view shows the basic information about the error codes, execute the following query to get a complete list of codes and their descriptions:

```
SELECT * FROM M_ERROR_CODES ORDER BY CODE ASC;
```

The *SAP HANA Troubleshooting and Performance Analysis Guide* has a section on Error Codes including a reference topic with links to additional information for many error codes (link in Related Information).

The following table lists error codes and their descriptions for the SAP HANA database:

Code	Type	Description
1	WRN_GENERAL	General warning
2	ERR_GENERAL	General error
3	FATAL_GENERAL	Fatal error
4	FATAL_OUT_OF_MEMORY	Cannot allocate enough memory
5	ERR_INIT	Initialization error
6	ERR_DATA	Invalid data
7	ERR_FEATURE_NOT_SUPPORTED	Feature not supported
8	ERR_INV_ARGUMENT	Invalid argument

Code	Type	Description
9	ERR_INDEX_OUT_OF_BOUNDS	Index out of bounds
10	ERR_AUTHENTICATION_FAILED	Authentication failed
11	ERR_INV_STATE	Invalid state
12	ERR_FILE_OPEN_FAILED	Cannot open file
13	ERR_FILE_CREATE_WRITE_FAILED	Cannot create/write file
14	ERR_DISK_SPACE_SHORTAGE	Cannot allocate enough disk space
15	ERR_FILE_NOT_FOUND	Cannot find file
16	ERR_RETRY_STATEMENT	Statement retry
17	ERR_CATA_VER_MISMATCH	Metadata schema version incompatible between database and executable
18	ERR_SERVICE_SHUTDOWN	Service shutting down
19	ERR_INV_LICENSE	Invalid license
20	ERR_CON_OUTSIDE_VALIDITY_PERIOD	Connect attempt outside user's validity period
21	ERR_PERSISTENCE	Persistence error
128	ERR_TX	Transaction error
129	ERR_TX_ROLLBACK	Transaction rolled back by an internal error
130	ERR_TX_ROLLBACK_INTEGRITY	Transaction rolled back by integrity constraint violation
131	ERR_TX_ROLLBACK_LOCK_TIMEOUT	Transaction rolled back by lock wait timeout
132	ERR_TX_ROLLBACK_RESOURCE	Transaction rolled back due to unavailable resource
133	ERR_TX_ROLLBACK_DEADLOCK	Transaction rolled back by detected deadlock
134	FATAL_REC_CHKPT_FILE_FAIL	Failure in accessing checkpoint file
135	FATAL_REC_ANCHOR_FILE_FAIL	Failure in accessing anchor file
136	FATAL_REC_LOG_FILE_FAIL	Failure in accessing log file
137	FATAL_REC_ARCHIVE_FILE_FAIL	Failure in accessing archive file
138	ERR_TX_SERIALIZATION	Transaction serialization failure
139	ERR_TX_ROLLBACK_QUERY_CANCEL	Current operation cancelled by user and transaction rolled back
140	ERR_TX_INV_TID	Invalid write-transaction identifier
141	ERR_INV_LOG_FILE_FAIL	Failure in accessing invisible log file

Code	Type	Description
142	ERR_TX_EXCEED_MAX_TX_NUM	Exceed max num of concurrent transactions
143	ERR_TX_SERIALIZATION_WITH_TIMEOUT	Transaction serialization failure until timeout expires
144	ERR_TX_ROLLBACK_UNIQUE_VIOLATED	Transaction rollback
145	ERR_TX_DIST_FAILURE	Transaction distribution work failure
146	ERR_TX_LOCK_ACQUISITION_FAIL	Resource busy and NOWAIT specification
147	ERR_TX_DATA_LOG_INCONSISTENT	Inconsistency between data and log
148	ERR_TX_START_BLOCKED	Transaction start is blocked until Master_Restart finishes
149	ERR_TX_DIST_2PC_FAILURE	Distributed transaction commit failure
150	ERR_TX_SNAPSHOT_TOO_OLD	Statement cancelled or snapshot timestamp already invalidated
151	ERR_TX_ROLLBACK_ROW_STORE_FULL	Transaction rollback
152	ERR_TX_ROLLBACK_ROW_STORE_FULL_DBAREQ	Transaction rollback
153	ERR_TX_SERIALIZATION_UNLOCK_REQUIRED	Transaction serialization failure
154	ERR_TX_INDEX_HANDLE_ACQUISITION_FAIL	Failure in acquiring index handle
155	ERR_TX_ROLLBACK_DEFERRED_FK	Transaction rollback
208	ERR_TX_XAER	XA transaction error
209	ERR_TX_XAER_ASYNC	Asynchronous operation error
210	ERR_TX_XAER_DUPID	Duplicate XID
211	ERR_TX_XAER_INVALID	Invalid argument of XA call
212	ERR_TX_XAER_NOTA	No valid XID
213	ERR_TX_XAER_OUTSIDE	Outside of global transaction
214	ERR_TX_XAER_PROTO	Improper XA call sequence
215	ERR_TX_XAER_RMERR	Resource manager error
216	ERR_TX_XAER_RMFAIL	Resource manager unavailable
217	ERR_TX_XA_RBPROTO	A protocol error occurred in the resource manager
218	ERR_TX_XA_RBROLLBACK	The rollback was caused by an unspecified reason
256	ERR_SQL	Sql processing error
257	ERR_SQL_PARSE	Sql syntax error
258	ERR_SQL_INSUFF_PRIV	Insufficient privilege

Code	Type	Description
259	ERR_SQL_INV_TABLE	Invalid table name
260	ERR_SQL_INV_COLUMN	Invalid column name
261	ERR_SQL_INV_INDEX	Invalid index name
262	ERR_SQL_INV_QUERY	Invalid query name
263	ERR_SQL_INV_ALIAS	Invalid alias name
264	ERR_SQL_INV_DATATYPE	Invalid datatype
265	ERR_SQL_MISSING_EXP	Expression missing
266	ERR_SQL_INCNST_DATATYPE	Inconsistent datatype
267	ERR_SQL_LONG_LEN_TYPE	Specified length too long for its d
268	ERR_SQL_AMBG_COLUMN	Column ambiguously defined
269	ERR_SQL_MANY_VALUES	Too many values
270	ERR_SQL_FEW_VALUES	Not enough values
271	ERR_SQL_DPLC_ALIAS	Duplicate alias
272	ERR_SQL_DPLC_COLUMN	Duplicate column name
273	ERR_SQL_LONG_CHAR	Not a single character string
274	ERR_SQL_INS_LARGE_VALUE	Inserted value too large for column
275	ERR_SQL_NOT_FUNCTION	Aggregate function not allowed
276	ERR_SQL_NOT_SINGLE_GROUP	Missing aggregation or grouping
277	ERR_SQL_NOT_GROUP_EXP	Not a GROUP BY expression
278	ERR_SQL_NESTED_WO_GROUP	Nested group function without Gf BY
279	ERR_SQL_TOO_DEEP_NESTED	Group function is nested
280	ERR_SQL_ORDER_EXCEED_NUM	ORDER BY item must be the num SELECT-list
281	ERR_SQL_OUTER_IN_OR	Outer join not allowed in operand or IN
282	ERR_SQL_OUTER_CROSS_JOIN	Two tables cannot be outer-joined other
283	ERR_SQL_OUTER_MORE_TWO	A table may be outer joined to at one other table
284	ERR_SQL_JOIN_NOT_MATCH	Join field does not match
285	ERR_SQL_INV_JOIN_PRED	Invalid join condition
286	ERR_SQL_LONG_IDENTIFIER	Identifier is too long

Code	Type	Description
287	ERR_SQL_NOT_NULL	Cannot insert NULL or update to
288	ERR_SQL_EXST_TABLE	Cannot use duplicate table name
289	ERR_SQL_EXST_INDEX	Cannot use duplicate index name
290	ERR_SQL_EXST_QUERY	Cannot use duplicate query name
291	ERR_SQL_NOT_POS_ARGUMENT	Argument identifier must be posi
292	ERR_SQL_FEW_ARGUMENT	Wrong number of arguments
293	ERR_SQL_INV_ARGUMENT	Argument type mismatch
294	ERR_SQL_MANY_PRIMARY_KEY	Cannot have more than one prim
295	ERR_SQL_LONG_MULTIKY	Too long multi key length
296	ERR_SQL_REP_TABLE_KEY	Replicated table must have a prin
297	ERR_SQL_REP_UPDATE_KEY	Cannot update primary key field i replicated table
298	ERR_SQL_NOT_DDL_STORE	Cannot store DDL
299	ERR_SQL_NOT_DROP_SYSIDX	Cannot drop index used for enfor of unique/primary key
300	ERR_SQL_ARG_OUT_OF_RANGE	Argument index is out of range
301	ERR_SQL_UNIQUE_VIOLATED	Unique constraint violated
302	ERR_SQL_INV_CHAR_VAL	Invalid CHAR or VARCHAR value
303	ERR_SQL_INV_DATETIME_VAL	Invalid DATE
304	ERR_SQL_DIV_BY_ZERO	Division by zero undefined
305	ERR_SQL_SINGLE_ROW	Single-row query returns more th row
306	ERR_SQL_INV_CURSOR	Invalid cursor
307	ERR_SQL_NUM_OUT_OF_RANGE	Numeric value out of range
308	ERR_SQL_EXST_COLUMN	Column name already exists
309	ERR_SQL_SUBQ_TOP_ORDERBY	Correlated subquery cannot have ORDER BY
310	ERR_SQL_IN_PROC	Sql error in procedure
311	ERR_SQL_DROP_ALL_COLUMNS	Cannot drop all columns in a tabl
312	ERR_SQL_SEQ_EXHAUST	Sequence is exhausted
313	ERR_SQL_INV_SEQ	Invalid sequence
314	ERR_SQL_OVERFLOW_NUMERIC	Numeric overflow
315	ERR_SQL_INV_SYNONYM	Invalid synonym

Code	Type	Description
316	ERR_SQL_INV_NUM_ARG_FUNC	Wrong number of arguments in function invocation
317	ERR_SQL_NOT_MATCH_PLAN_TABLE	""P_QUERYPLANS"" not exists in table format
318	ERR_SQL_DECIMAL_PRECISION	Decimal precision specifier is out of range
319	ERR_SQL_DECIMAL_SCALE	Decimal scale specifier is out of range
320	ERR_SQL_LOB_INDEX	Cannot create index on expression of datatype LOB
321	ERR_SQL_INV_VIEW	Invalid view name
322	ERR_SQL_EXST_VIEW	Cannot use duplicate view name
323	ERR_SQL_REP_DPLC_ID	Duplicate replication id
324	ERR_SQL_EXST_SEQ	Cannot use duplicate sequence name
325	ERR_SQL_ESC_SEQ	Invalid escape sequence
326	ERR_SQL_SEQ_CURRVAL	CURRVAL of given sequence is not defined in this session
327	ERR_SQL_CANNOT_EXPLAIN	Cannot explain plan of given statement
328	ERR_SQL_INV_FUNC_PROC	Invalid name of function or procedure
329	ERR_SQL_EXST_FUNC_PROC	Cannot use duplicate name of function or procedure
330	ERR_SQL_EXST_SYNONYM	Cannot use duplicate synonym name
331	ERR_SQL_EXST_USER	User name already exists
332	ERR_SQL_INV_USER	Invalid user name
333	ERR_SQL_COLUMN_NOT_ALLOWED_HERE	Column not allowed
334	ERR_SQL_INV_PRIV	Invalid user privilege
335	ERR_SQL_EXST_ALIAS	Field alias name already exists
336	ERR_SQL_INV_DEFAULT	Invalid default value
337	ERR_SQL INTO NOT ALLOWED	INTO clause not allowed for this SQL statement
338	ERR_SQL_ZERO_LEN_NOT_ALLOWED	Zero-length columns are not allowed
339	ERR_SQL_INV_NUMBER	Invalid number
340	ERR_SQL_VAR_NOT_BOUND	Not all variables bound
341	ERR_SQL_UNDERFLOW_NUMERIC	Numeric underflow
342	ERR_SQL_COLLATE_CONFLICT	Collation conflict

Code	Type	Description
343	ERR_SQL_INV_COLLATE_NAME	Invalid collate name
344	ERR_SQL_LOADER_PARSE	Parse error in data loader
345	ERR_SQL_NOT_REP_TABLE	Not a replication table
346	ERR_SQL_INV_REP_ID	Invalid replication id
347	ERR_SQL_INV_OPTION	Invalid option
348	ERR_SQL_INV_DATETIME_FORMAT	Invalid datetime format
349	ERR_SQL_CREATE_UNIQUE_INDEX	"Cannot CREATE UNIQUE INDEX; duplicate key found"
350	ERR_SQL_DROP_COL_PRIMARY_KEY	Cannot drop columns in the primary key column list
351	ERR_SQL_DROP_MULTI_COL_UNIQUE	Column is referenced in a multi-column constraint
352	ERR_SQL_CREATE_UNIQUE_INDEX_ON_CDX_TAB	Cannot create unique index on cd
353	ERR_SQL_EXST_UPDATE_LOG_GROUP	Update log group name already e
354	ERR_SQL_INV_UPDATE_LOG_GROUP_NAME	Invalid update log group name
355	ERR_SQL_UPDATE_LOG_TABLE_KEY	The base table of the update log table must have a primary key
356	ERR_SQL_MAX_UPDATE_LOG_GROUP	Exceed maximum number of update log group
357	ERR_SQL_BASE_TABLE_ALREADY_HAS_UL	The base table already has a update log table
358	ERR_SQL_UL	Update log table can not have a update log table
359	ERR_SQL_STR_LENGTH_TOO_LARGE	String is too long
360	ERR_SQL_VIEW_CHECK_VIOLATION	View WITH CHECK OPTION where violation
361	ERR_SQL_VIEW_UPDATE_VIOLATION	Data manipulation operation not allowed on this view
362	ERR_SQL_INV_SCHEMA	Invalid schema name
363	ERR_SQL_MAX_NUM_INDEX_COLUMN	Number of index columns exceed maximum
364	ERR_SQL_INV_PARTIAL_KEY_SIZE	Invalid partial key size
365	ERR_SQL_NO_MATCHING_UNIQUE_OR_PRIMARY_KEY	No matching primary key for this list
366	ERR_SQL_NO_PRIMARY_KEY	Referenced table does not have a primary key

Code	Type	Description
367	ERR_SQL_MISMATCH_OF_COLUMN_NUMBERS	Number of referencing columns do not match referenced columns
368	ERR_SQL_TEMP_TABLE_WITH_UNIQUE	Unique constraint not allowed on temporary table
369	ERR_SQL_MAX_VIEW_DEPTH	Exceed maximum view depth limit
370	ERR_SQL_DIRECT_INSERT_WITH_UNIQUE_INDEX	Cannot perform DIRECT INSERT operation on table with unique index
371	ERR_SQL_XML_PARSE	Invalid XML document
372	ERR_SQL_XPATH_PARSE	Invalid XPATH
373	ERR_SQL_INV_XML_DURATION	Invalid XML duration value
374	ERR_SQL_INV_XML_FUNCTION	Invalid XML function usage
375	ERR_SQL_INV_XML_INDEX_OPERATION	Invalid XML index operation
376	ERR_SQL_PYTHON	Python builtin procedure error
377	ERR_SQL_JIT	JIT operation error
378	ERR_SQL_INV_COLUMN_VIEW	Invalid column view
379	ERR_SQL_TABLE_SCHEMA_MISMATCH	Table schema mismatch
380	ERR_SQL_RUN_LEVEL_CHANGE	Fail to change run level
381	ERR_SQL_RESTART	Fail to restart
382	ERR_SQL_COLLECT_ALL_VERSIONS	Fail to collect all version garbage
383	ERR_SQL_INV_IDENTIFIER	Invalid identifier
384	ERR_SQL_TOO_LONG_CONSTANT	String is too long
385	ERR_SQL_RESTORE_SESSION	Could not restore session
386	ERR_SQL_EXST_SCHEMA	Cannot use duplicate schema name
387	ERR_SQL_AMBG_TABLE	Table ambiguously defined
388	ERR_SQL_EXST_ROLE	Role already exists
389	ERR_SQL_INV_ROLE	Invalid role name
390	ERR_SQL_INV_USERTYPE	Invalid user type
391	ERR_SQL_INV_USABLE_VIEW	Invalidated view
392	ERR_SQL_CYCLIC_ROLES	Can't assign cyclic role
393	ERR_SQL_NO_GRANT_OPTION_FOR_ROLE	Roles must not receive a privilege grant option
394	ERR_SQL_CANT_REVOKE_ROLE	Error revoking role
395	ERR_SQL_INV_USER_DEFINED_TYPE	Invalid user-defined type name

Code	Type	Description
396	ERR_SQL_EXST_USER_DEFINED_TYPE	Cannot use duplicate user-define name
397	ERR_SQL_INV_OBJ_NAME	Invalid object name
398	ERR_SQL_MANY_ORDER_BY	Cannot have more than one order
399	ERR_SQL_TOO_DEEP_ROLE_TREE	Role tree too deep
400	ERR_SQL_INSERT_ONLY_TABLE_WITH_PRIMARY_KEY	Primary key not allowed on insert table
401	ERR_SQL_INSERT_ONLY_TABLE_WITH_UNIQUE	Unique constraint not allowed on only table
402	ERR_SQL_DROPPED_USER	The user was already dropped before query execution
403	ERR_SQL_INTERNAL_ERROR	Internal error
404	ERR_SQL_INV_STRUCTURED_PRIVILEGE_NAME	Invalid (non-existent) structured privilege name
405	ERR_SQL_DUP_STRUCTURED_PRIVILEGE_NAME	Cannot use duplicate structured privilege name
406	ERR_SQL_CANT_UPDATE_GEN_COL	INSERT
407	ERR_SQL_INV_DATE_FORMAT	Invalid date format
408	ERR_SQL_PASS_OR_PARAMETER_NEEDED	Password or parameter required for query
409	ERR_SQL_TOO_MANY_PARAMETER_VALUES	Multiple values for a parameter not supported
410	ERR_SQL_INV_PRIVILEGE_NAMESPACE	Invalid privilege namespace
411	ERR_SQL_INV_TABLE_TYPE	Invalid table type
412	ERR_SQL_INV_PASSWORD_LAYOUT	Invalid password layout
413	ERR_SQL_PASSWORD_REUSED	Last n passwords can not be reused
414	ERR_SQL_ALTER_PASSWORD_NEEDED	User is forced to change password
415	ERR_SQL_USER_DEACTIVATED	User is deactivated
416	ERR_SQL_USER_LOCKED	"User is locked; try again later"
417	ERR_SQL_CANT_DROP_WITHOUT_CASCADE	Can't drop without CASCADE specification
418	ERR_SQL_INV_VIEW_QUERY	Invalid view query for creation
419	ERR_SQL_CANT_DROP_WITH_RESTRICT	Can't drop with RESTRICT specification
420	ERR_SQL_ALTER_PASSWORD_NOT_ALLOWED	Password change currently not allowed
421	ERR_SQL_FULLTEXT_INDEX	Cannot create fulltext index

Code	Type	Description
422	ERR_SQL_MIXED_PRIVILEGE_NAMESPACES	Privileges must be either all SQL c from one namespace
423	ERR_SQL_LVC	AFL error
424	ERR_SQL_INV_PACKAGE	Invalid name of package
425	ERR_SQL_EXST_PACKAGE	Duplicate package name
426	ERR_SQL_NUM_COLUMN_MISMATCH	Number of columns mismatch
427	ERR_SQL_CANT_RESERVE_INDEX_ID	Can not reserve index id any more
428	ERR_INV_QUERY_PLAN_ID	Invalid query plan id
429	ERR_SQL_INTEGRITY_CHECK_FAILED	Integrity check failed
430	ERR_SQL_INV_USABLE_PROC	Invalidated procedure
431	WRN_SQL_NEARLY_EXPIRED_PASSWORD	User's password will expire within days
432	WRN_SQL_DEPRECATED_SYNTAX	This syntax has been deprecated be removed in next release
433	ERR_SQL_NOT_NULL_CONSTRAINT	Null value found
434	ERR_SQL_INV_OBJECT	Invalid object ID
435	ERR_SQL_INV_EXP	Invalid expression
436	ERR_SQL_SET_SYSTEM_LICENSE	Could not set system license
437	ERR_SQL_ONLY_LICENSE_HANDLING	Only commands for license handl allowed in current state
438	ERR_SQL_INVALID_USER_PARAMETER_VALUE	Invalid user parameter value
439	ERR_SQL_COMPOSITE_ERROR	Composite error
440	ERR_SQL_TABLE_TYPE_CONVERSION_ERROR	Table type conversion error
441	WRN_SQL_DEPRECATED_FEATURE	This feature has been deprecated be removed in next release
442	ERR_SQL_MAX_NUM_COLUMN	Number of columns exceeds its maximum
443	ERR_SQL_INV_CALC_SCENARIO	Invalid calculation scenario name
444	ERR_SQL_PACKMAN	Package manager error
445	ERR_SQL_INV_TRIGGER	Invalid trigger name
446	ERR_SQL_EXST_TRIGGER	Cannot use duplicate trigger nam
447	ERR_SQL_BACKUP_FAILED	Backup could not be completed
448	ERR_SQL_RECOVERY_FAILED	Recovery could not be completed

Code	Type	Description
449	ERR_SQL_RECOVERY_STRATEGY	Recovery strategy could not be determined
450	ERR_SQL_UNSET_SYSTEM_LICENSE	Failed to unset system license
451	ERR_SQL_NOT_ALLOWED_SUBJ_TAB_ACCESS_TRIGGER	Modification of subject table in tr not allowed
452	ERR_SQL_INV_BACKUPID	Invalid backup id
453	ERR_SQL_USER_WITHOUT_PASSWORD	User does not have a password
454	WRN_SQL_WRONG_HINT_SYNTAX	Wrong hint syntax
455	ERR_SQL_READ_ONLY_SESSION_VARIABLE	The predefined session variable c be set via SET command
456	ERR_SQL_NOT_ALLOWED_FOR_SPECIAL_ROLE	Not allowed for this role
457	ERR_SQL_DPLC_CONSTRAINT	Duplicate constraint name
458	ERR_SQL_UNSUPPORTED_FUNCTION	Unsupported function included
459	ERR_SQL_INV_USABLE_FUNC	Invalidated function
460	ERR_SQL_INV_PRIVILEGE_FOR_OBJECT	Invalid privilege for object
461	ERR_SQL_FK_NOT_FOUND	Foreign key constraint violation
462	ERR_SQL_FK_ON_UPDATE_DELETE_FAILED	Failed on update or delete by fore constraint violation
463	ERR_SQL_MAX_NUM_TABLE	Number of tables exceeds its ma
464	ERR_SQL_MAX_PARSE_TREE_DEPTH	SQL internal parse tree depth exc maximum
465	ERR_SQL_INV_USABLE_TRIGGER	Cannot execute trigger
466	ERR_SQL_CREDENTIAL_NOT_FOUND	No credential found
467	ERR_SQL_PARAM_VARIABLE	Cannot use parameter variable
468	ERR_SQL_HINT	Hint error
469	ERR_SQL_INV_SRC_DATATYPE	Unsupported datatype on source
470	ERR_SQL_INV_DATA_SOURCE_CONF	Invalid data source configuration
471	ERR_SQL_INV_DATA_SOURCE	Invalid data source name
472	ERR_SQL_EXST_DATA_SOURCE	Cannot use duplicate data source
473	ERR_SQL_ADAPTER_CONFIGURATION	Invalid adapter configuration
474	ERR_SQL_INV_ADAPTER	Invalid adapter name
475	ERR_SQL_EXST_ADAPTER	Cannot use duplicate adapter nar
476	ERR_SQL_INV_REMOTE_OBJECT	Invalid remote object name

Code	Type	Description
477	ERR_SQL_CREDENTIAL_EXISTS	Credential exists
478	ERR_SQL_UDF_RUNTIME	User defined function runtime err
479	ERR_SQL_INV_SPATIAL_ATTRIBUTE	Invalid spatial attribute
480	ERR_SQL_INV_SPATIAL_UNIT	Invalid spatial unit of measure na
481	ERR_SQL_EXST_SPATIAL_UNIT	Cannot use duplicate spatial unit measure name
482	ERR_SQL_INV_SPATIAL_REF_SYS	Invalid spatial reference system n
483	ERR_SQL_EXST_SPATIAL_REF_SYS	Cannot use duplicate spatial refer system name
484	ERR_SQL_SESSION_GROUP_COMMAND_FAILURE	Invalid session group command
485	ERR_SQL_INV_STRUCTURED_PRIVILEGE_DEFINITION	Invalid definition of structured pri
486	WRN_SQL_IMPORT_PARTIALLY_FAILED	Some of rows have failed to be im
487	ERR_SQL_IMPORT_PARTIALLY_FAILED	Some of rows have failed to be im
488	ERR_SQL_INV_DATABASE	Invalid database name
489	ERR_SQL_INV_EPMMODEL	Invalid EPM Model name
490	ERR_SQL_EXST_EPMMODEL	Cannot use duplicate EPM Model
491	ERR_SQL_INV_EPMMODEL_DEF	Invalid EPM Model definition
492	ERR_SQL_INV_EPMQUERYSOURCE	Invalid EPM Query Source name
493	ERR_SQL_EXST_EPMQUERYSOURCE	Cannot use duplicate EPM Query name
494	ERR_SQL_INV_EPMQUERYSOURCE_DEF	Invalid EPM Query Source definiti
495	ERROR_SQL_INVALID_CONV_TO_EXTENDED	Table already Extended with right option
496	ERROR_SQL_INVALID_CONV_FROM_EXTENDED	Table already Non-Extended
497	ERROR_EXTENDED_STORAGE_NOT_FOUND	Extended Storage not found for ta
498	ERR_SQL_IMPORT_FAIL_ON_MAX_RECORD_SIZE_CHECK	Memory for a record exceeds the
499	ERR_SQL_INV_C2C	Invalid stacked column search
500	ERR_SQL_REQUIRE_PREDICATE	Predicates are required in a where
501	ERR_SQL_SERIES_INVALID_SPEC	Invalid series data specification:
502	ERR_SQL_INV_TASK	Invalid name of task
503	ERR_SQL_EXST_TASK	Cannot use duplicate name of tas
504	ERR_SQL_INV_ADAPTER_LOCATION	Invalid adapter location
505	ERR_SQL_LAST_ADAPTER_LOCATION	Cannot remove last location of ac

Code	Type	Description
506	ERR_SQL_SYSTEM_ADAPTER	Invalid create
507	ERR_SQL_INV_AGENT	Invalid agent name
508	ERR_SQL_EXST_AGENT	Cannot use duplicate agent name
509	ERR_SQL_INV_AGENT_PROPS	Invalid agent properties
510	ERR_SQL_TEMP_TABLE_IN_USE	Cannot alter global temporary table use or create/alter/drop index on table
512	ERR_REP	Replication error
513	ERR_SQL_REP_ALREADY_ACTIVE	Cannot execute DDL statement on replication table while replicating
514	FATAL_REP_ANCHOR_FILE_FAIL	Failure in accessing anchor file
515	FATAL_REP_LOG_FILE_FAIL	Failure in accessing log file
516	ERR_REP_TABLE_HAVE_NOT_CONFLICT_TABLE	Replication table has not conflict table
517	ERR_REPORT_TABLE_ALREADY_ENABLED	Conflict report table already enabled
518	ERR_REPORT_TABLE_ALREADY_DISABLED	Conflict report table already disabled
544	ERR_RS_PARTITION	Partition error
545	ERR_RS_PARTITION_INVALID_SPEC	Invalid partition spec
546	ERR_RS_PARTITION_INVALID_ID	Invalid partition id
547	ERR_RS_PARTITION_INVALID_PROPERTY	Invalid use of partition property
548	ERR_RS_PARTITION_INVALID_RANGE	Invalid partition range
549	ERR_RS_PARTITION_INVALID_TYPE	Invalid partition type
550	ERR_RS_PARTITION_INVALID_COLUMN	Invalid partition column
551	ERR_RS_PARTITION_INVALID_LEVEL	Invalid partition level
576	ERR_API	Api error
577	ERR_API_NO_CURSOR_FWD	Cursor type of forward is not allowed
578	ERR_API_INV_STATEMENT	Invalid statement
579	ERR_API_EXCEED_MAX_GROUP_SIZE	Exceed maximum batch size
580	ERR_API_VERSION_INCOMPATIBLE	Server rejected the connection(protocol version mismatch)
581	ERR_API_ONLY_SINGLE_STATEMENT	This function can be called only in case of single statement
582	ERR_API_INV_RETURN_OBJECT	This query does not have result set
583	ERR_API_CONNECTION_DOES_NOT_EXIST	Connection does not exist

Code	Type	Description
584	ERR_API_NO_MORE_LOB_DATA	No more LOB data
585	ERR_API_OPERATION_NOT_PERMITTED	Operation is not permitted
586	ERR_API_INV_PARAMETER_FROM_SERVER	Invalid parameter is received from
587	ERR_API_INV_RESULTSET	Result set is currently invalid
588	ERR_API_RESULTSET_NEXT_NOT_CALLED	Next() is not called for this result
589	ERR_API_TOO_MANY_PARAMETERS	Too many parameters are set
590	ERR_API_MISSING_PARAMETER	Some paramters are missing
591	ERR_API_INTERNAL_ERROR	Internal error
592	ERR_API_NOT_SUPPORTED_TYPECONV	Not supported type conversion
593	ERR_API_REMOTE_ONLY	Remote-only function
594	ERR_API_NO_MORE_RESULT_ROW	No more result row in result set
595	ERR_API_NOT_OUT_PARAM	Specified parameter is not output parameter
596	ERR_API_LOB_STREAM_NOT_PERMITTED	LOB streaming is not permitted in commit mode
597	ERR_API_SESSION_CONTEXT_ERROR	Session context error
598	ERR_API_EXTERNAL_EXECUTION_FAILURE	Failed to execute the external statement
599	ERR_API_NOT_INITIALIZED	Session layer is not initialized yet
600	ERR_API_CALL_ROUTING_FAILURE	Failed routed execution
601	ERR_API_TOO_MANY_SESSION_VARIABLES	Too many session variables are set
602	ERR_API_READONLY_SESSION_VARIABLE	Cannot set read-only session variable
603	ERR_API_INV_LOB	Invalid LOB
604	ERR_API_REMOTE_TEMP_TABLE	Remote temp table access failure
605	ERR_API_INV_XA_JOIN	Invalid xa join request
606	ERR_API_EXCEED_MAX_LOB_SIZE	Exceed maximum LOB size
607	ERR_API_CLEANUP_RESOURCE	Failed to cleanup resources
608	ERR_API_EXCEED_MAX_PREPARED_STATEMENT	Exceed maximum number of prepared statements
609	ERR_API_INVALID_CESU8_STRING	Invalid CESU-8 string
610	ERR_API_ROLLBACK_FAILURE	Transaction rollback failure
611	ERR_API_SESSION_VARIABLE_VALUE_LENGTH_EXCEEDED	Maximum length of value for session variable exceeded

Code	Type	Description
612	ERR_API_SESSION_VARIABLE_KEY_LENGTH_EXCEEDED	Maximum length of key for session variable exceeded
613	ERR_API_TIMEOUT	Execution aborted by timeout
614	ERR_API_ACCESS_ENCRYPTED_DATA	Encrypted data access is not permitted
615	ERR_API_REMOTE_CONNECTION_DOES_NOT_EXIST	Remote connection does not exist
616	ERR_API_REJECTED_BY_WORKLOAD_CLASS	Rejected by workload class configuration
640	ERR_SQL_2	Sql processing error
641	ERR_SQL_INV_REMOTE_SUBSCRIPTION	Invalid remote subscription name
642	ERR_SQL_EXST_REMOTE_SUBSCRIPTION	Cannot use duplicate remote subscription name
643	ERR_SQL_INV_REMOTE_SUBSCRIPTION_DEF	Invalid remote subscription definition
644	ERR_SQL_EXST_REMOTE_SOURCE_ADAPTER_LOCATION	Remote source refers to the adapter location
645	ERR_SQL_EXST_REMOTE_SOURCE_ACTIVE_SUBSCRIPTIONS	Remote source has active remote subscriptions:
646	ERR_SQL_INV_USABLE_TASK	Invalidated task
647	ERR_SQL_NOT_ALLOWED_SYNTAX_FOR_TRIGGER	Not supported syntax in trigger
648	ERR_SQL_TRIGGER_AND_PROC_NESTING_DEPTH_EXCEEDED	Nesting depth of trigger and procedure exceeded
649	ERR_SQL_QUERY_PINNED_PLAN	Pinned plan error
650	ERR_SQL_QUERY_REMOVE_PINNED_PLAN	Remove pinned plan error
651	ERR_SQL_EXST_OBJECT	Cannot use duplicate object name
652	ERR_SQL_AMBG_SCHEMA	Schema ambiguously defined
653	ERR_SQL_SET_ROW_ORDER	Row order already set on table
654	ERR_SQL_NO_ROW_ORDER	No row order on table set
655	ERR_SQL_LICENSING_RUNTIME	Licensing error
656	ERR_SQL_LONG_PROPERTY	Property value too long
657	ERR_SQL_CANCEL_TASK_TIMEOUT_REACHED	Request to cancel task was sent but did not cancel before timeout was reached
658	ERR_SQL_CANNOT_MUTATE_TABLE_DURING_FK_EXECUTION	Cannot mutate the table during trigger foreign key execution
659	ERR_SQL_EXST_WORKLOAD_CLASS	Cannot use duplicate workload class name
660	ERR_SQL_INV_WORKLOAD_CLASS	Invalid workload class name

Code	Type	Description
661	ERR_SQL_EXST_WORKLOAD_MAPPING	Cannot use duplicate workload m name
662	ERR_SQL_INV_WORKLOAD_MAPPING	Invalid workload mapping name
663	ERR_SQL_CONNECT_NOT_ALLOWED	User not allowed to connect from
664	ERR_SQL_INV_AGENT_GROUP	Invalid agent group name
665	ERR_SQL_EXST_AGENT_GROUP	Cannot use duplicate agent group
666	ERR_SQL_AGENT_GROUP_NOT_EMPTY	Agents are still set to this agent g
667	ERR_SQL_TEXT_MINING_FAILURE	Text mining error
668	ERR_SQL_2D_POINTS_SUPPORTED_ONLY	ST_Point columns support 2-dim points only
669	ERR_SQL_SPATIAL_ERROR	Spatial error
670	ERR_SQL_PART_NOT_EXIST	Part does not exist
671	ERR_SQL_EXST_LIBRARY	Cannot use duplicate library nam
672	ERR_SQL_DPLC_ASSOCIATION	Duplicate association name
673	ERR_SQL_INV_GRAPH_WORKSPACE	Invalid graph workspace name
674	WRN_SQL_EXPORT_SKIP_CROSSDB_OBJECT	Cross database object found & sk exporting
675	ERR_SQL_EXST_GRAPH_WORKSPACE	Cannot use duplicate graph work: name
676	ERR_SQL_DUP_WORKLOAD_MAPPING	Cannot use duplicate combinatio workload mapping
677	ERR_SQL_CHECK_CONSTRAINT_VIOLATION	Check constraint violation
678	ERR_SQL_PLANSTABILIZER_DEPRECATED	Deprecated plan stabilizer error
679	ERR_SQL_PLANSTABILIZER_NO_MANAGER_DEPRECATED	Deprecated plan stabilizer error - manager not found: please check Stabilizer is enabled
680	ERR_SQL_STATEMENT_HINT	Statement hint error
681	ERR_SQL_STATEMENT_HINT_COMMAND	Statement hint error - error while processing statement hint comm
682	ERR_SQL_STATEMENT_HINT_TABLE_EMPTY	Statement hint error - statement table is empty
683	ERR_SQL_STATEMENT_HINT_MAP_LOAD_ERROR	Statement hint error - statement table is corrupt
684	ERR_SQL_STATEMENT_HINT_RECORD_ALREADY_EXISTS	Statement hint error - statement record already exists

Code	Type	Description
685	ERR_SQL_STATEMENT_HINT_RECORD_DOES_NOT_EXIST	Statement hint error - statement record does not exist
686	ERR_SQL_START_TASK_ERROR	Start task error
687	ERR_SQL_EXCEED_LAG_TIME	Exceed lag time of RESULT_LAG
688	ERR_IO_FAILURE_ON_FILE_WRITE	I/O error occurred on file write
689	ERR_SQL_DUPLICATE_ROWID_MATCHED	Duplicate rowid matched during r into
690	ERR_SQL_PLAN_STABILITY_DEPRECATED	Deprecated plan stability error
691	ERR_SQL_PLAN_STABILITY_COMMAND_DEPRECATED	Deprecated plan stability error - e while processing command
692	ERR_SQL_PLAN_STABILITY_TABLE_EMPTY_DEPRECATED	Deprecated plan stability error - s plan table is empty
693	ERR_SQL_PLAN_STABILITY_MAP_LOAD_ERROR_DEPRECATED	Deprecated plan stability error - s plan table is corrupt
694	ERR_SQL_PLAN_STABILITY_RECORD_ALREADY_EXISTS_DEPRECATED	Deprecated plan stability error - s plan record already exists
695	ERR_SQL_PLAN_STABILITY_RECORD_DOES_NOT_EXIST_DEPRECATED	Deprecated plan stability error - s plan record does not exist
696	ERR_SQL_PLAN_STABILITY_CANNOT_CONVERT_ABSTRACT_PLAN_DEPRECATED	Deprecated plan stability error - c convert to abstract plan
697	ERR_SQL_PREACTIVE_KEY_EXISTS	Preactive key already exists
698	ERR_SQL_NO_PREACTIVE_KEY	No preactive key exists
699	ERR_SQL_EXST_DEPENDENCY_RULE	Cannot use duplicate dependency name
700	ERR_SQL_SINGLE_COLUMN_SEARCH_THROW_ERROR	No_stacked_column_search(thro error
701	ERR_SQL_EXST_USERGROUP	Usergroup name already exists
702	ERR_SQL_INV_USERGROUP	Invalid usergroup name
703	ERR_INCORRECT_ROOT_KEYS_BACKUP_PASSWORD	Incorrect root keys backup password
704	ERR_SQL_USERGROUP_DELETION_FAILED	Usergroup cannot be dropped
705	ERR_SQL_CONCURRENT_GRANT	Two concurrent statements perform the same grant operation
706	ERR_SQL_INV_SYMMETRIC_CIPHER	Currently only AES-256-CBC is supported: invalid cipher
707	ERR_SQL_EXST_COLUMN_KEY	Cannot use duplicate column key
708	ERR_SQL_EXST_COLUMN_KEYCOPY	Column keycopy already exists

Code	Type	Description
709	ERR_SQL_EXST_KEYPAIR	Keypair already exists
710	ERR_SQL_INV_ASYMMETRIC_CIPHER	Currently only RSA-OAEP-2048 is supported: invalid cipher
711	ERR_SQL_EXST_COLUMN_KEY_ID	Cannot use duplicate column key
712	ERR_SQL_PLAN_STABILITY_MIGRATION_DEPRECATED	Deprecated plan stabilizer stored error - migration error
713	ERR_SQL_NOT_OWN_KEYPAIR	Keypair not owned by the creator column key
714	ERR_SQL_DROP_COLUMN_KEYCOPY	Cannot drop the last key admin key
715	ERR_SQL_EMPTY_WORKLOAD_MAPPING	Cannot use a workload mapping with properties
716	ERR_SQL_STALE_STATEMENT	Statement is stale
717	ERR_SQL_INV_KEY_ID	Invalid key id
718	ERR_SQL_CANNOT_UPDATE_NOT_EXISTING_ROW	Cannot update non-existent row
719	ERR_SQL_CURRENCY_UNIT_CONVERSION	Currency/unit conversion error
720	ERR_SQL_DDL_DISABLED	DDL is currently disabled in this session
721	ERR_SQL_MISSING_ROWID	Rowid column is not found
722	ERR_SQL_DROP_LAST_COLUMN_KEYCOPY	Cannot drop the last keycopy
723	ERR_SQL_NOT_KEY_ADMIN_KEYPAIR	Keypair not owned by key admin
724	ERR_SQL_INV_LIBRARY	Invalid name of library
725	ERR_SQL_INV_USABLE_LIBRARY	Invalidated library
726	ERR_SQL_INVALID_BASE64_STRING	Invalid Base64 string
727	ERR_SQL_INV_CACHE	Invalid result cache name
728	ERR_SQL_EXST_CACHE	Cannot use duplicate result cache
729	ERR_SQL_UNKNOWN_SESSION	Unknown session id
730	ERR_SQL_RECOMPILE_WITHOUT_FALLBACK	Query recompilation is required for reoptimized plan generation
731	ERR_SQL_RECOMPILE_WITH_FALLBACK	Query recompilation is required for legacy plan generation
732	ERR_SQL_FAIL_AUTO_MATCH_CACHE	Cannot find matching view
733	ERR_SQL_DDL_ONLY_OBJECT	No DML on DDL-only object
734	ERR_SQL_EMPTY_COLUMN_KEY	Cannot use empty column encryption
735	ERR_SQL_RECOMPILE_WITH_DISK	Query recompilation is required for plan generation

Code	Type	Description
736	ERR_SQL_REMOVE_SQL_PLAN_CACHE_SELECTIVE	Invalid WHERE condition.
737	ERR_SQL_FUZZY_SEARCH_INDEX	Cannot create fuzzy search index
738	ERR_SQL_INV_RESULT_CACHE_TYPE	Invalid result cache type
1024	ERR_SES	Session error
1025	ERR_COM	Communication error
1026	ERR_COM_LISTEN	Cannot bind a communication po
1027	ERR_COM_INIT	Communication initialization erro
1028	ERR_COM_IOCTL	I/O control error
1029	ERR_COM_CONNECT	Connection failure
1030	ERR_COM_SEND	Send error
1031	ERR_COM_RECEIVE	Receive error
1032	ERR_SES_THREAD_CREATE	Cannot create a thread
1033	ERR_SES_INV_PROTOCOL	Error while parsing protocol
1034	ERR_SES_EXCEED_MAX_SESSION	Exceed maximum number of sess
1035	ERR_SES_INV_VERSION	Not supported version
1036	ERR_SES_INV_SESSION	Invalid session id
1037	ERR_COM_UNKNOWN_HOST	Unknown hostname
1038	ERR_SES_SERVER_BUSY	Rejected as server is temporarily overloaded
1088	ERR_DATA_STAT	Data statistics error
1089	ERR_DATA_STAT_NOT_FOUND	No matching data statistics objec found
1090	ERR_DATA_STAT_REMOTE_QUERY_ERR	Invalid result from query to remot source
1091	ERR_DATA_STAT_TABLE_NOT_FOUND	Specified table not found or not supported
1092	ERR_DATA_STAT_BUILD_ERROR	Error building data statistics obje
1093	ERR_DATA_STAT_EXISTS	Data statistics object already exis
1094	ERR_DATA_STAT_INVALID_SETTING	Invalid combination of settings sp
1120	ERR_USERGROUP_GENERAL	Usergroup error
1121	ERR_USERGROUP_USER_NOT_MEMBER_OF_ANY	User is not member of any usergr
1122	ERR_USERGROUP_EQUAL_CURRENT_AND_NEW_USERGROUP	Current and new usergroup are th
1123	ERR_USERGROUP_UNKNOWN_PARAMETER_NAME	Unknown parameter for usergrou

Code	Type	Description
1124	ERR_USERGROUP_UNKNOWN_PARAMETER_SET_NAME	Unknown parameter set for usergroup
1125	ERR_USERGROUP_DUPLICATE_PARAMETER_NAME	Same parametername specified more than once
1126	ERR_USERGROUP_INVALID_PARAMETER_VALUE	Invalid value for usergroup parameter
1127	ERR_USERGROUP_DISABLE_CONNECT_BY_GROUP_MEMBER_NOT_ALLOWED	User is member of this usergroup
1152	ERR_SQL_PLAN_STABILITY	Plan stability error
1153	ERR_SQL_PLAN_STABILITY_COMMAND	Plan stability error - cannot execute command
1154	ERR_SQL_PLAN_STABILITY_TABLE_EMPTY	Plan stability error - abstract sql plan table is empty
1155	ERR_SQL_PLAN_STABILITY_MAP_LOAD_ERROR	Plan stability error - cannot load abstract sql plan
1156	ERR_SQL_PLAN_STABILITY_RECORD_ALREADY_EXISTS	Plan stability error - abstract sql plan record already exists
1157	ERR_SQL_PLAN_STABILITY_RECORD_DOES_NOT_EXIST	Plan stability error - abstract sql plan record does not exist
1158	ERR_SQL_PLAN_STABILITY_CANNOT_CAPTURE	Cannot capture abstract sql plan
1159	ERR_SQL_PLAN_STABILITY_MIGRATION	Plan stability error - migration
1160	ERR_SQL_PLAN_STABILITY_CANNOT_APPLY	Cannot apply abstract sql plan
1161	ERR_SQL_PLAN_STABILITY_JSON_ERROR	Cannot deserialize abstract sql plan
1162	ERR_SQL_PLAN_STABILITY_UPDATE_LOCATION	Plan stability error - update location
1163	ERR_SQL_PLAN_STABILITY_RELATED_OBJECT_CHECK_FAILED	Plan stability error - related object mismatch
1280	ERR_SQLSCRIPT_2	Sqlscript error
1281	ERR_SQLSCRIPT_WRONG_PARAMS	Wrong number or types of parameter call
1282	ERR_SQLSCRIPT_OUT_PARAM_VAR	Output parameter not a variable
1283	ERR_SQLSCRIPT_OUT_PARAM_DEFAULT	OUT and IN OUT parameters may have default expressions
1284	ERR_SQLSCRIPT_DUP_PARAMETERS	Duplicate parameters are not permitted
1285	ERR_SQLSCRIPT_DUP_DECL	At most one declaration is permitted in the declaration section
1286	ERR_SQLSCRIPT_CURSOR_SELECT_STMT	Cursor must be declared by SELECT statement
1287	ERR_SQLSCRIPT_ID_NOT_DECLARED	Identifier must be declared

Code	Type	Description
1288	ERR_SQLSCRIPT_NOT_ASSIGN_TARGET	Expression cannot be used as an assignment target
1289	ERR_SQLSCRIPT_NOT_INTO_TARGET	Not allowed expression for INTO-clause
1290	ERR_SQLSCRIPT_LHS_CANNOT_ASSIGNED	Expression is inappropriate as the left hand side of an assignment statement
1291	ERR_SQLSCRIPT_EXPR_WRONG_TYPE	Expression is of wrong type
1292	ERR_SQLSCRIPT_ILLEGAL_EXIT_STMT	Illegal loop control statement
1293	ERR_SQLSCRIPT_ID_EXCEPTION_TYPE	Not a condition variable
1294	ERR_SQLSCRIPT_INTO_CLAUSE	An INTO clause is expected in SELECT statement
1295	ERR_SQLSCRIPT_NOT_ALLOWED_STMT	Not permitted statement
1296	ERR_SQLSCRIPT_NOT_CURSOR	Identifier is not a cursor
1297	ERR_SQLSCRIPT_NUM_FETCH_VALUES	Wrong number of values in the IN clause of a FETCH statement
1298	ERR_SQLSCRIPT_UNHANDLED_EXCEPTION	Unhandled user-defined exception
1299	ERR_SQLSCRIPT_NO_DATA_FOUND	No data found
1300	ERR_SQLSCRIPT_FETCH_MANY_ROWS	Fetch returns more than requested number of rows
1301	ERR_SQLSCRIPT_VALUE_ERROR	Numeric or value error
1302	ERR_SQLSCRIPT_OUT_PARAM_IN_FUNCTION	Parallelizable function cannot have an OUT parameter
1303	ERR_SQLSCRIPT_USER_DEFINED_EXCEPTION	User-defined exception
1304	ERR_SQLSCRIPT_CURSOR_ALREADY_OPEN	Cursor is already opened
1305	ERR_SQLSCRIPT_INVALID_RETURN_TYPE	Return type is invalid
1306	ERR_SQLSCRIPT_RETURN_TYPE_MISMATCH	Return type mismatch
1307	ERR_SQLSCRIPT_UNSUPPORTED_DATATYPE	Unsupported datatype is used
1308	ERR_SQLSCRIPT_INVALID_SINGLE_ASSIGNMENT	Illegal single assignment
1309	ERR_SQLSCRIPT_INVALID_USE_OF_TABLE_VARIABLE	Invalid use of table variable
1310	ERR_SQLSCRIPT_NOT_ALLOWED_SCALAR_TYPE	Scalar type is not allowed
1311	ERR_SQLSCRIPT_NO_OUT_PARAM	Out parameter is not specified
1312	ERR_SQLSCRIPT_AT_MOST_ONE_OUT_PARAM	At most one output parameter is allowed
1313	ERR_SQLSCRIPT_OUT_PARAM_TABLE	Output parameter should be a table or table variable
1314	ERR_SQLSCRIPT_INVALID_VARIABLE_NAME	Invalid or reserved variable name

Code	Type	Description
1315	ERR_SQLSCRIPT_RETURN_RESULT_SET_WITH_RESULTVIEW	Return result set from SELECT statement exist when result view is defined
1316	ERR_SQLSCRIPT_NOT_ASSIGNED_OUT_TABVAR	Some out table variable is not assigned
1317	ERR_SQLSCRIPT_FUNCTION_NAME_MAX_LEN	Function name exceeds maximum limitation
1318	ERR_SQLSCRIPT_BUILTIN_NOT_DEFINED	Built-in function not defined
1319	ERR_SQLSCRIPT_BUILTIN_PARAM_NOT_TABLE_NAME	Parameter must be a table name
1320	ERR_SQLSCRIPT_BUILTIN_PARAM_ATTRIBUTE_WITH_SCHEMA	Parameter must be an attribute name without a table name upfront
1321	ERR_SQLSCRIPT_BUILTIN_PARAM_ATTRIBUTE_WITH_ALIAS	Parameter must be an attribute name without an alias
1322	ERR_SQLSCRIPT_CALC_ATTR_NOT_ALLOWED	CE_CALC not allowed
1323	ERR_SQLSCRIPT_BUILTIN_PARAM_NOT_COL_OR_AGGR_VECTOR	Parameter must be a vector of column or aggregations
1324	ERR_SQLSCRIPT_BUILTIN_MISSING_JOIN_ATTR_IN_PROJECTION	Join attribute must be available in projection list
1325	ERR_SQLSCRIPT_BUILTIN_PARAM_NOT_SQLIDENT_VECTOR	Parameter must be a vector of sql identifiers
1326	ERR_SQLSCRIPT_DUPLICATE_ATTRIBUTE_NAME	Duplicate attribute name
1327	ERR_SQLSCRIPT_PARAM_UNSUPPORTED_TYPE	Parameter has a non supported type
1328	ERR_SQLSCRIPT_BUILTIN_MISSING_ATTRIBUTE_IN_PROJECTION	Attribute not found in column table
1329	ERR_SQLSCRIPT_BUILTIN_DUPLICATE_COLUMN_NAME	Duplicate column name
1330	ERR_SQLSCRIPT_BUILTIN_CALCATTR_EXPRESSION_SYNTAX	Syntax Error for calculated Attribute
1331	ERR_SQLSCRIPT_BUILTIN_FILTER_EXPRESSION_SYNTAX	Syntax Error in filter expression
1332	ERR_SQLSCRIPT_BUILTIN_FIRST_PARAM_NOT_COLUMN_TABLE	Parameter must be a valid column or projection view on column table
1333	ERR_SQLSCRIPT_BUILTIN_JOINATTR_NOT_FOUND_IN_VAR	Join attributes not found in variable
1334	ERR_SQLSCRIPT_BUILTIN_IN_PARAM_NOT_SAME_TABLE_TYPE	Input parameters do not have the same table type
1335	ERR_SQLSCRIPT_RUNTIME_CYCLIC_DEPENDENCY	Cyclic dependency found in a runtime procedure
1336	ERR_SQLSCRIPT_RUNTIME_UNEXPECTED_EXCEPTION	Unexpected internal exception caught in a runtime procedure
1337	ERR_SQLSCRIPT_VAR_DEPENDS_ON_UNASSIGNED_VAR	Variable depends on an unassigned variable
1338	ERR_SQLSCRIPT_CE_CONVERSION_CUSTOM_TAB_MISSING	Cannot find table used in CE_CONVERSION

Code	Type	Description
1339	ERR_SQLSCRIPT_TOO_MANY_PARAMS	Too many parameters
1340	ERR_SQLSCRIPT_NESTED_CALL_TOO_DEEP	The depth of the nested call is too
1341	ERR_SQLSCRIPT_VERSION_VALIDATION_FAILED	Procedure version validation failed
1342	ERR_SQLSCRIPT_CE_CALC_ATTRIBUTE_AND_ALIAS_ARE_SAME	Attribute has the same name as the
1343	ERR_SQLSCRIPT_RETRY_EXCEPTION	Retry Exception is occurred in a procedure
1344	ERR_SQLSCRIPT_NOT_ALLOWED_DYNAMIC_SQL	Dynamic SQL or DDL is not allowed
1345	ERR_SQLSCRIPT_NOT_ALLOWED_CONCURRENT_WRITES	Not allowed concurrent write
1346	ERR_SQLSCRIPT_NOT_ALLOWED_CONCURRENT_READ_AND_WRITE	Not allowed concurrent read and
1347	WRN_SQLSCRIPT_NOT_RECOMMENDED_FEATURE	Not recommended feature
1348	ERR_SQLSCRIPT_LLANG_GET_LIBRARY_IMPORT_LIST_FAILED	Failed to retrieve the list of imported libraries from LLANG procedure
1349	ERR_SQLSCRIPT_INITIAL_ASSIGNMENT_REQUIRED_FOR_CONSTANT_TABLE	Assigning initial value is required for declaring constant table variable
1350	ERR_SQLSCRIPT_NOT_ALLOWED_NON_DETERMINISTIC_FEATURE	Non-deterministic feature is not allowed
1351	ERR_SQLSCRIPT_INVALID_PARSE_TREE	Invalid parse tree
1352	ERR_SQLSCRIPT_ENCRYPTION_NOT_ALLOWED	Not allowed for encrypted procedure function
1353	ERR_SQLSCRIPT_NOT_NULL_COLUMN_IGNORED	NOT NULL constraints in explicit function types are ignored
1354	ERR_SQLSCRIPT_CURSOR_NOT_OPENED	Cursor to be fetched has not been opened yet
1355	ERR_SQLSCRIPT_INVALID_EXTERN_LANG	Invalid external language
1356	ERR_SQLSCRIPT_COMPOSITE	Composite error in SQLScript procedure
1357	ERR_SQLSCRIPT_CE_TYPE_MISMATCH	Datatype mismatch in CE function
1358	ERR_SQLSCRIPT_LIB_WITHOUT_USING	Require USING statement before
1359	ERR_SQLSCRIPT_TOO_MANY_MEMBERS	Surpassed expected library member count
1360	ERR_SQLSCRIPT_INVALID_INPUT	Invalid input
1361	ERR_SQLSCRIPT_INVALID_PRAGMA	Invalid pragma
1362	ERR_SQLSCRIPT_CALLSTACK_TOO_DEEP	The depth of the sqlscript callstack is too deep
1363	ERR_SQLSCRIPT_USER_CONDITION	User-defined condition
1364	ERR_SQLSCRIPT_ASYNC_CALL_INIT_FAILED	Asynchronous procedure call initialization failed

Code	Type	Description
1365	ERR_SQLSCRIPT_ASYNC_CALL_CANCELLATION_FAILED	Asynchronous procedure call cancellation failed
1536	ERR_STATISTICS_MONITOR_INTERNAL_ERROR	System monitor internal error
1776	ERR_CONFIG_UNSUPPORTED	Configuration error:
1777	WRN_CONFIG_UNSUPPORTED	Configuration warning:
1792	ERR_SHM	Shared memory error
1793	ERR_SHM_CREATE_INVALID	Invalid key or invalid size
1794	ERR_SHM_CREATE_ALREADY_EXIST	The segment already exists
1795	ERR_SHM_CREATE_EXCEED_LIMIT	Exceed the system-wide limit on s memory
1796	ERR_SHM_CREATE_NOT_EXIST	No segment exists for the given k
1797	ERR_SHM_CREATE_NO_ACCESS	The user does not have permissic access the shared memory segm
1798	ERR_SHM_CREATE_NO_MORE_MEMORY	No memory could be allocated fo segment overhead
1799	ERR_SHM_DROP_INVALID	Invalid shmid
1800	ERR_SHM_DROP_NO_ACCESS	Allow read access for shmid
1801	ERR_SHM_DROP_REMOVED_IDENTIFIER	Shmid points to a removed identi
1802	ERR_SHM_DROP_NO_PERMISSION	The effective user ID of the calling process is not the creator
1803	ERR_SHM_DROP_OVERFLOW	The gid or uid value is too large to stored in the structure
1804	ERR_SHM_ATTACH_NO_ACCESS	The user does not have permissic access the shared memory segm
1805	ERR_SHM_ATTACH_INVALID	Invalid shmid
1806	ERR_SHM_ATTACH_NO_MORE_MEMORY	No memory could be allocated fo descriptor or for the page tables
1807	ERR_SHM_UNKNOWN	Unknown shared memory error
1856	ERR_SQL_PLAN_EXECUTION_MONITOR_GENERAL	[SQL Plan Execution Monitor] ger error
2048	ERR_CS	Column store error
2049	ERR_CS_NO_PRIMARY_KEY	Primary key is not specified for cc table
2050	ERR_CS_NOT_SUPPORTED_DDL	Not supported DDL type for colur
2051	ERR_CS_NOT_SUPPORTED_DATA_TYPE	Not supported data type for colur
2052	ERR_CS_NOT_SUPPORTED_DML	Not supported DML type for colur

Code	Type	Description
2053	ERR_CS_INVALID_RETURNED_VALUE	Invalid returned value from attribute engine
2054	ERR_CS_DELTA_LOG_REPLAY_FAILED	Delta log replay failed
2055	ERR_CS_MAXIMUM_ROW	Maximum number of rows per table partition reached
2056	ERR_CS_CONSISTENCY_CHECK_FAILED	Column store table consistency check failed
2304	ERR_PYDBAPI	Python dbapi error
2305	ERR_PYDBAPI_INTERFACE_FAILURE	Interface failure
2306	ERR_PYDBAPI_PROGRAMMING_MISTAKE	Programming mistake
2307	ERR_PYDBAPI_INVALID_QUERY_PARAMETER	Invalid query parameter
2308	ERR_PYDBAPI_NOT_SUPPORTED_STR_ENCODING	Not supported encoding for string
2560	ERR_METADATA	Metadata error
2561	ERR_DIST_METADATA	Distributed metadata error
2562	ERR_DIST_METADATA_REDIRECT_FAILURE	DDL redirect error
2563	ERR_DIST_METADATA_DDL_NOTIFICATION_FAILURE	DDL notification error
2564	ERR_DIST_METADATA_INVALID_CONTAINER_ID	DDL invalid container id
2565	ERR_DIST_METADATA_INVALID_INDEX_ID	DDL invalid index id
2566	ERR_DIST_METADATA_TNSCLIENT_FAILURE	Distributed environment error
2567	ERR_DIST_METADATA_NETWORK_FAILURE	Network error
2568	ERR_DIST_METADATA_NOT_SUPPORT	Metadata update not supported in distributed
2569	ERR_DIST_METADATA_MASTER_UPDATE_FAILURE	Metadata update in master index failed
2570	ERR_METADATA_ROW_PAGE_INTEGRITY_FAILURE	Row page consistency check is failed
2571	ERR_INTEGRITY_BROKEN_METADATA	Integrity check detects broken metadata
2572	ERR_INTEGRITY_INCONSISTENCY	Integrity check detects inconsistency
2573	ERR_INTEGRITY_ORPHANED	Integrity check detects orphaned
2574	ERR_METADATA_SEGMENT_MIGRATION_GENERAL	Metadata segment migration preparation error
2575	ERR_METADATA_SEGMENT_MIGRATION_FATAL	Metadata segment migration interruption error
2576	ERR_CATALOG_INVALID_ADDRESS	Attempt to access metadata of invalid address (already deleted or corrupted)
2577	ERR_CATALOG_NULL_TERMINATION	Catalog object has wrong null termination

Code	Type	Description
2578	ERR_CATALOG_LOCATION_ERROR	Catalog object has wrong location
2579	ERR_CATALOG_INVALID_REFERENCE	Catalog object has invalid reference(corrupted address)
2580	ERR_CATALOG_UNEXPECTED_REFERENCE	Catalog object has wrong referen
2581	ERR_CATALOG_VALUE_DOMAIN_GENERAL	The value of catalog object is bro
2582	ERR_CATALOG_FOREIGN_KEY_CONSTRAINT	Can not find catalog object
2583	ERR_CATALOG_WRONG_VAR_SLOT_SIZE	Catalog object has wrong slot size
2584	ERR_CATALOG_CYCLIC_DEPENDENCY	Can not make cyclic dependency
2585	ERR_ES_BROKEN_METADATA	Inconsistency between HANA and catalog
2586	ERR_ES_METADATA_FIX_FAIL	Repair operation to fix inconsistency failed
2816	ERR_SQLSCRIPT	SqlScript Error
2817	ERR_SQLSCRIPT_BUILTIN_TOO_MANY_RETURN_PARAM	SqlScript Builtin Function
2818	ERR_SQLSCRIPT_FUNCTION_NOT_FOUND	SqlScript
2819	ERR_SQLSCRIPT_TEMPLATE_PARAMETER_NUMBER_WRONG	SqlScript
2820	ERR_SQLSCRIPT_VARIABLE_NOT_DECLARED	SqlScript
2821	ERR_SQLSCRIPT_DUPLICATE_VARIABLE_NAME	SqlScript
2822	ERR_SQLSCRIPT_SQL_EXECUTION_FAILED	SqlScript
2823	ERR_SQLSCRIPT_DROP_FUNCTION_FAILED	SqlScript
2824	ERR_SQLSCRIPT_LOAD_FUNCTION_FAILED	SqlScript
2825	ERR_SQLSCRIPT_SIGNATURE_MISMATCH_WITH_CATALOG	SqlScript
2826	ERR_SQLSCRIPT_REGISTER_FUNCTION_IN_CATALOG_FAILED	SqlScript
2827	ERR_SQLSCRIPT_SCALAR_INPUT_PARAMS_NOT_SUPPORTED	SqlScript
2828	ERR_SQLSCRIPT_LANGUAGE_NOT_SUPPORTED	SqlScript
2829	ERR_SQLSCRIPT_DROP_FUNCTION_FAILED_EXISTING_CALLER	SqlScript
2830	ERR_SQLSCRIPT_LLANG_EXACTLY_ONE_OUTPUT_PARAM	SqlScript
2831	ERR_SQLSCRIPT_BUILTIN_FIRST_PARAM_NOT_COLUMN_TABLE	SqlScript
2832	ERR_SQLSCRIPT_BUILTIN_PARAM_COUNT_NOT_IN_RANGE	SqlScript
2833	ERR_SQLSCRIPT_BUILTIN_PARAM_COUNT_MISMATCH	SqlScript
2834	ERR_SQLSCRIPT_BUILTIN_PARAM_NOT_INPUT	SqlScript
2835	ERR_SQLSCRIPT_BUILTIN_PARAM_NOT_TABLE_NAME	SqlScript